colour — First Iter — Last Iter

Fixed Gear AI Fishery | Fixed Gear BS Fishery | Fixed Gear GOA Fishery | Trawl Fishery

Fishery Ages

Fishery Lengths

Mean Composition Bin

25
20
15
10
5

65
60
55

1990 2000 2010 2020

Year

colour — First Iter — Last Iter

Japan LL Survey | NOAA LL AI Survey | NOAA LL BS Survey | NOAA LL GOA Survey

Survey Ages

Survey Lengths

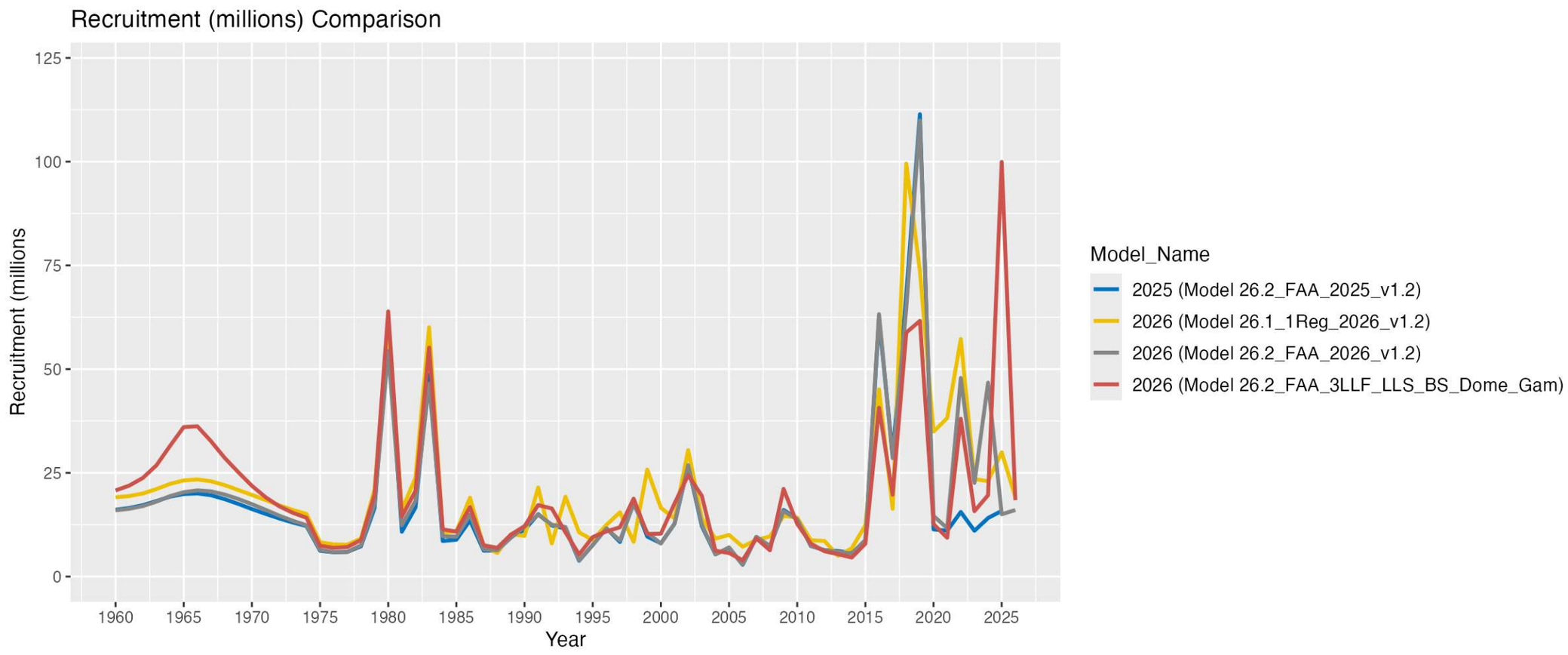
Mean Composition Bin

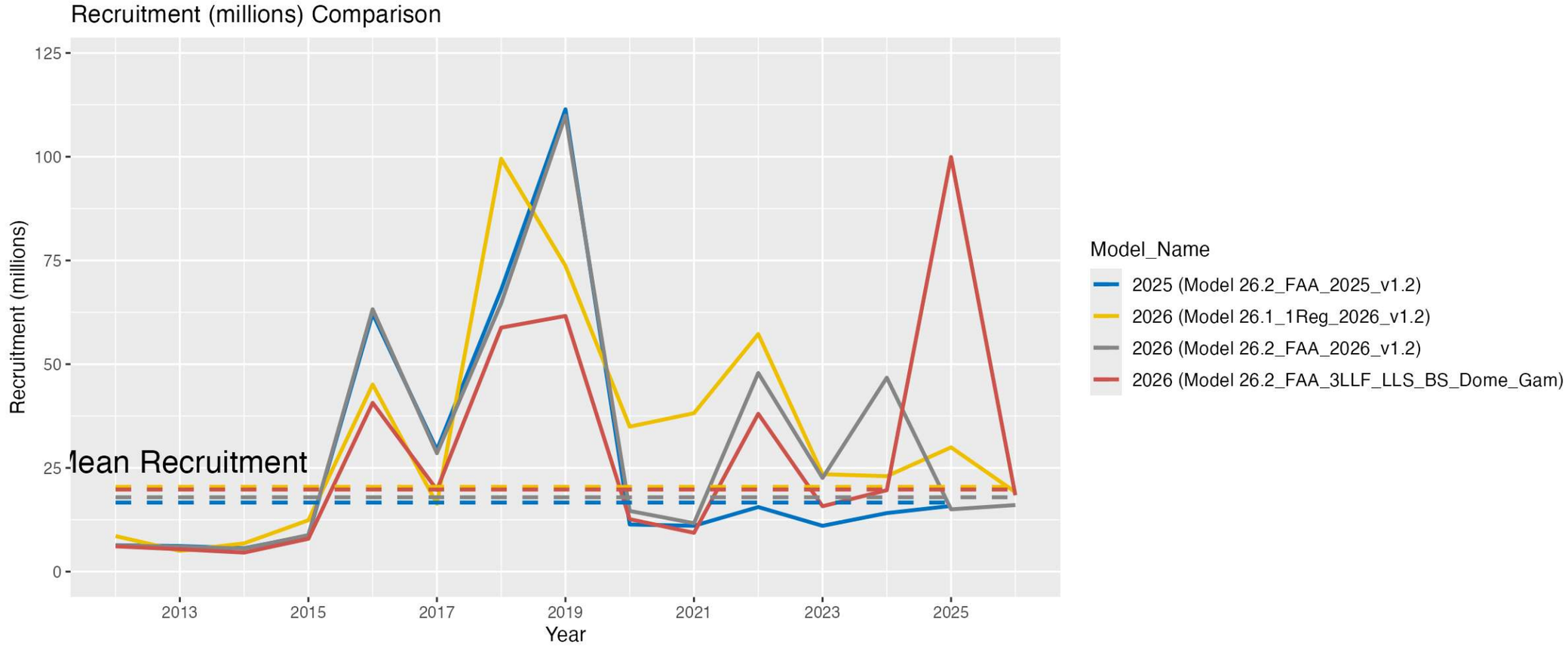
10.0
7.5
5.0

67.5
65.0
62.5

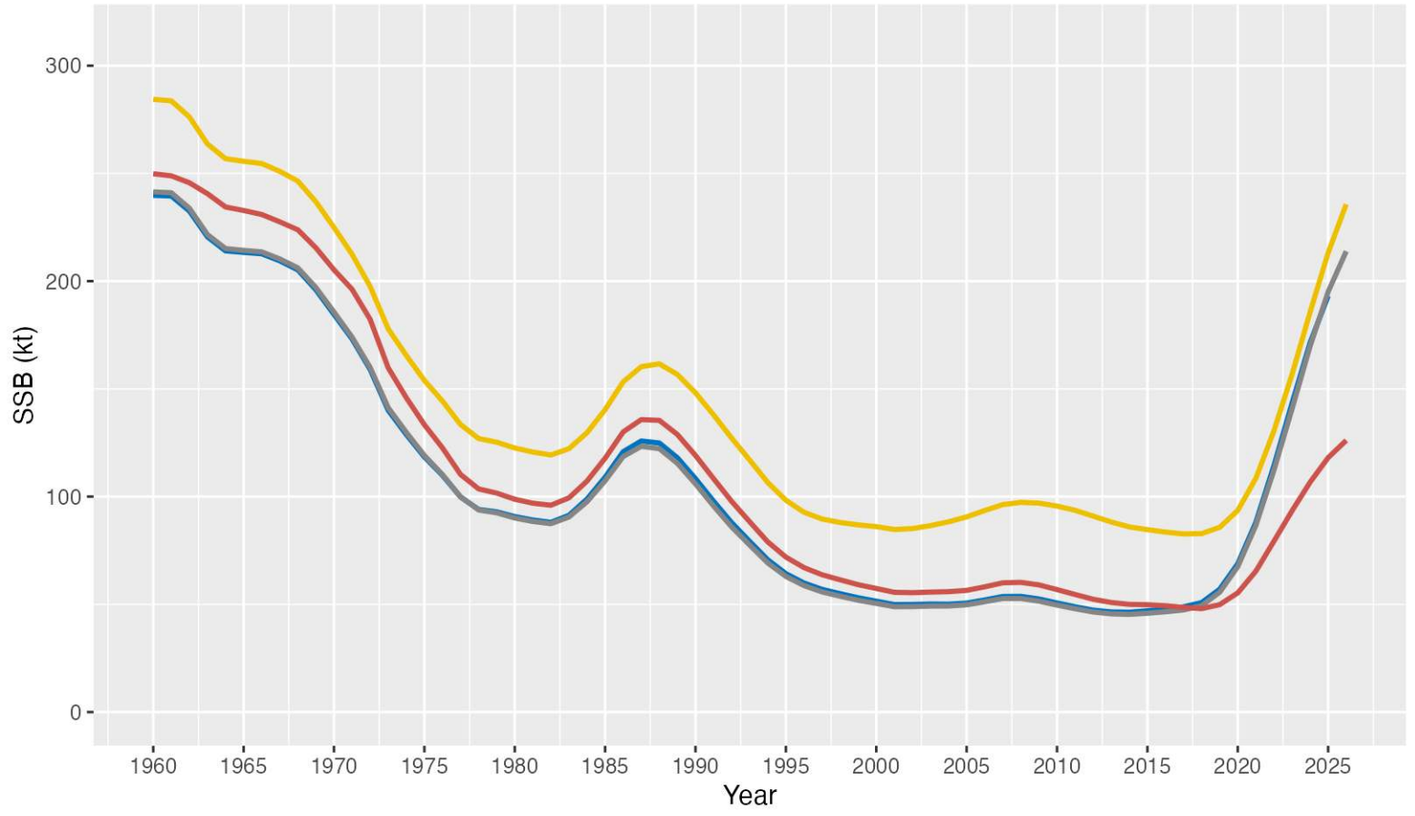
1980 1990 2000 2010 2020

Year


Recruitment (millions) Comparison
Recruitment (millions)
125
100
75
50
25
0
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
2025
Year
Model_Name
2025 (Model 26.2_FAA_2025_v1.2)
2026 (Model 26.1_1Reg_2026_v1.2)
2026 (Model 26.2_FAA_2026_v1.2)
2026 (Model 26.2_FAA_3LLF_LLS_BS_Dome_Gam)


Recruitment (millions) Comparison
Recruitment (millions)
125
100
75
50
25
0
lean Recruitment
2013
2015
2017
2019
2021
2023
2025
Year
Model_Name
2025 (Model 26.2_FAA_2025_v1.2)
2026 (Model 26.1_1Reg_2026_v1.2)
2026 (Model 26.2_FAA_2026_v1.2)
2026 (Model 26.2_FAA_3LLF_LLS_BS_Dome_Gam)

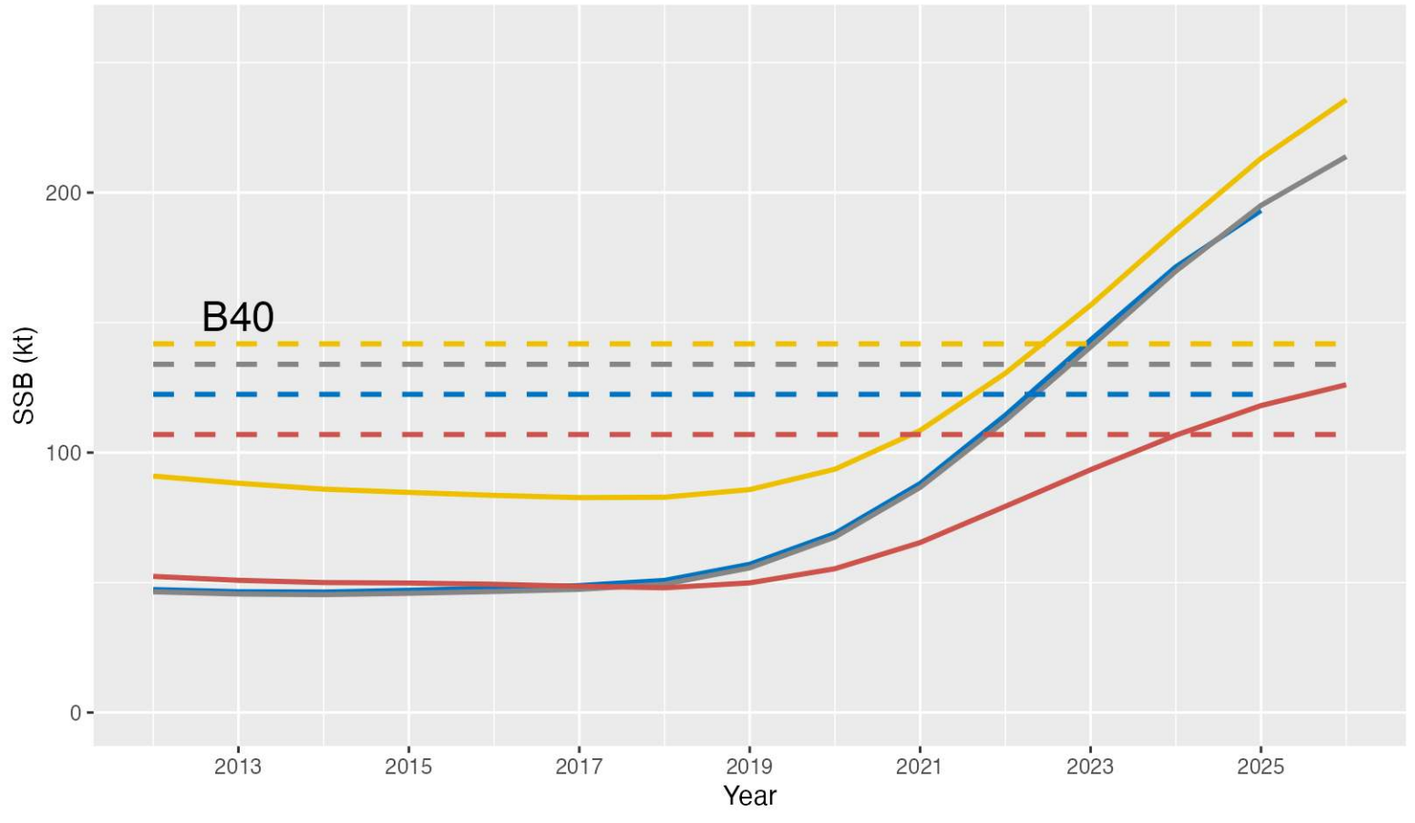
SSB (kt) Comparison



Model_Name

- 2025 (Model 26.2_FAA_2025_v1.2)
- 2026 (Model 26.1_1Reg_2026_v1.2)
- 2026 (Model 26.2_FAA_2026_v1.2)
- 2026 (Model 26.2_FAA_3LLF_LLS_BS_Dome_Gam)

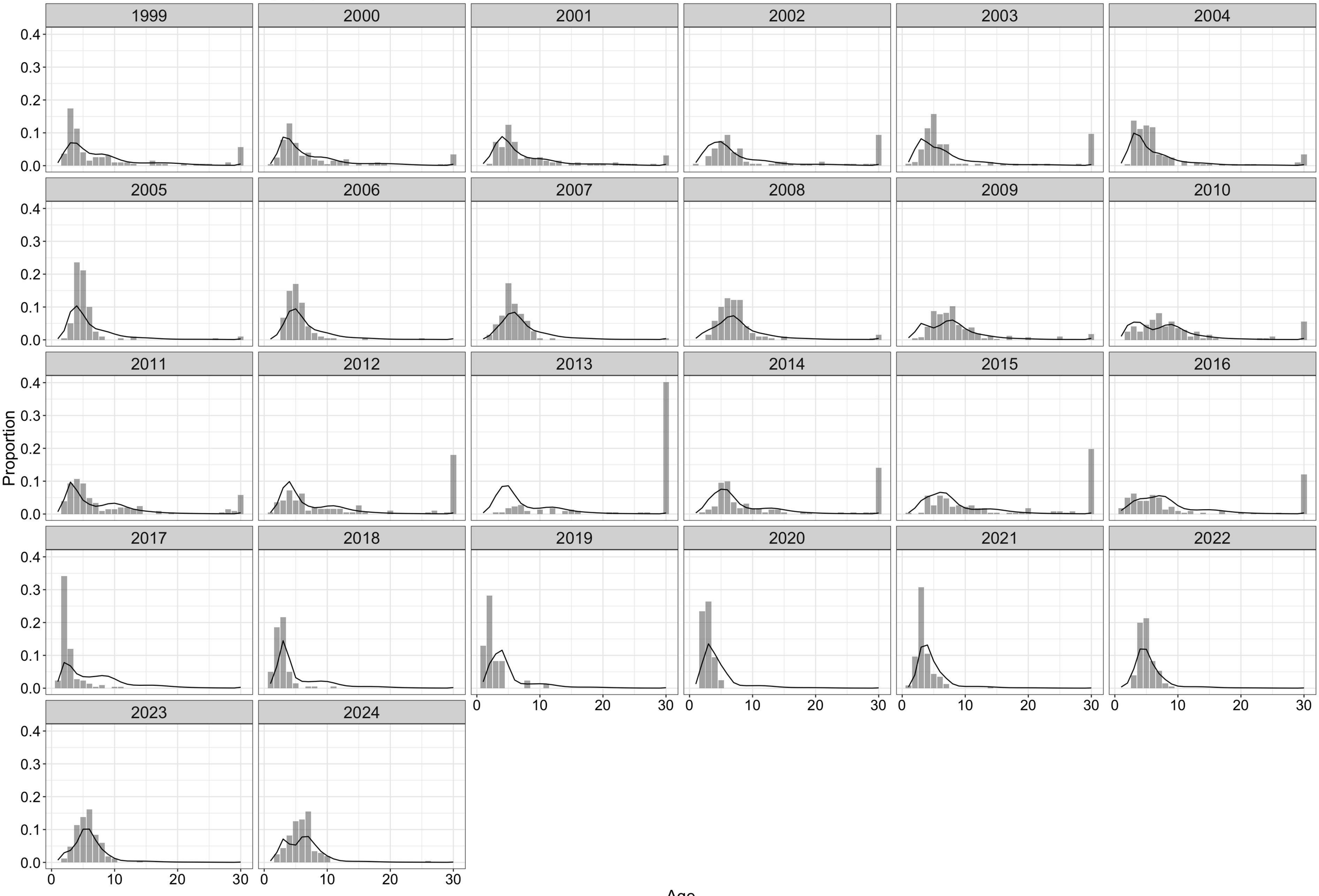
SSB (kt) Comparison



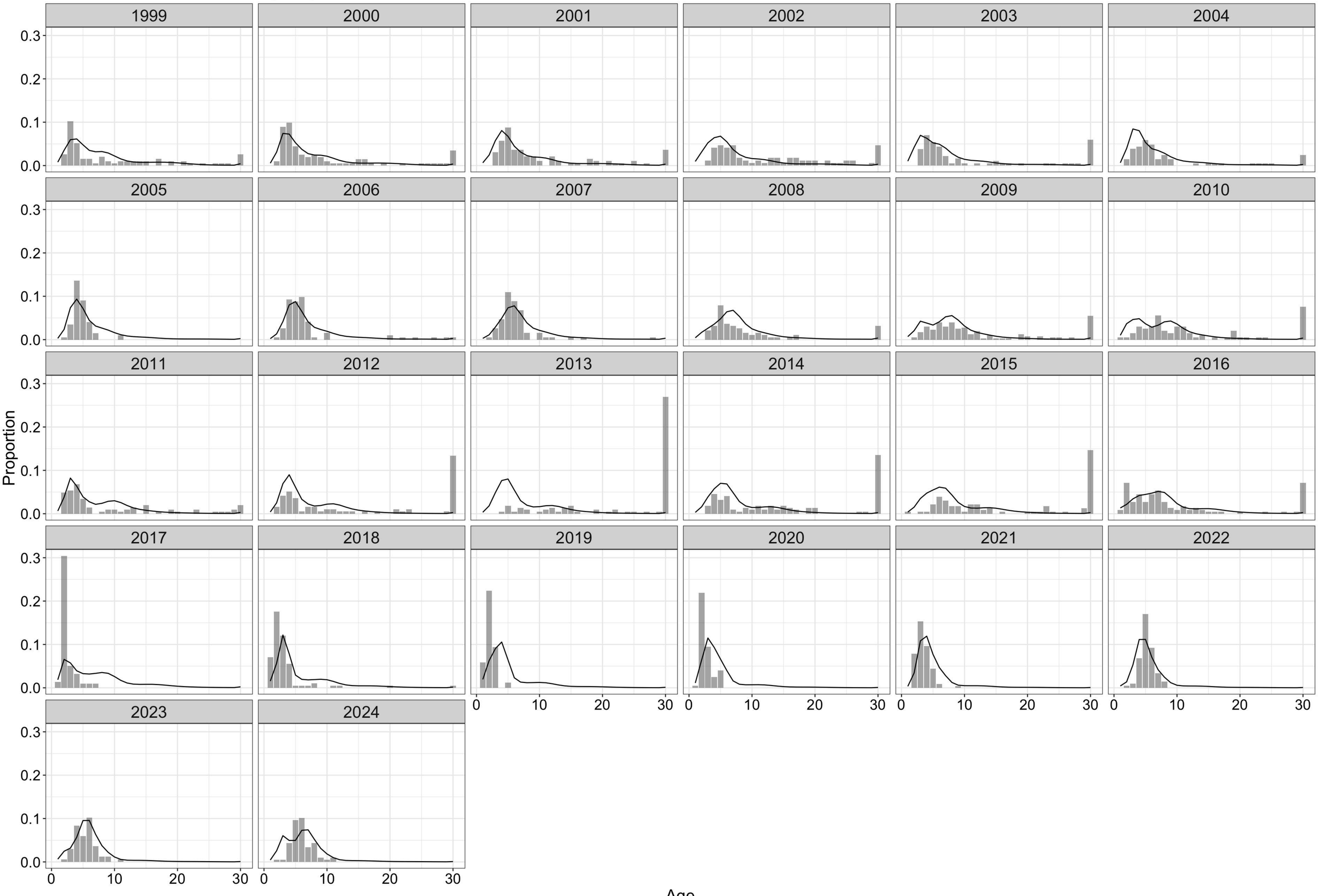
Model_Name

- 2025 (Model 26.2_FAA_2025_v1.2)
- 2026 (Model 26.1_1Reg_2026_v1.2)
- 2026 (Model 26.2_FAA_2026_v1.2)
- 2026 (Model 26.2_FAA_3LLF_LLS_BS_Dome_Gam)

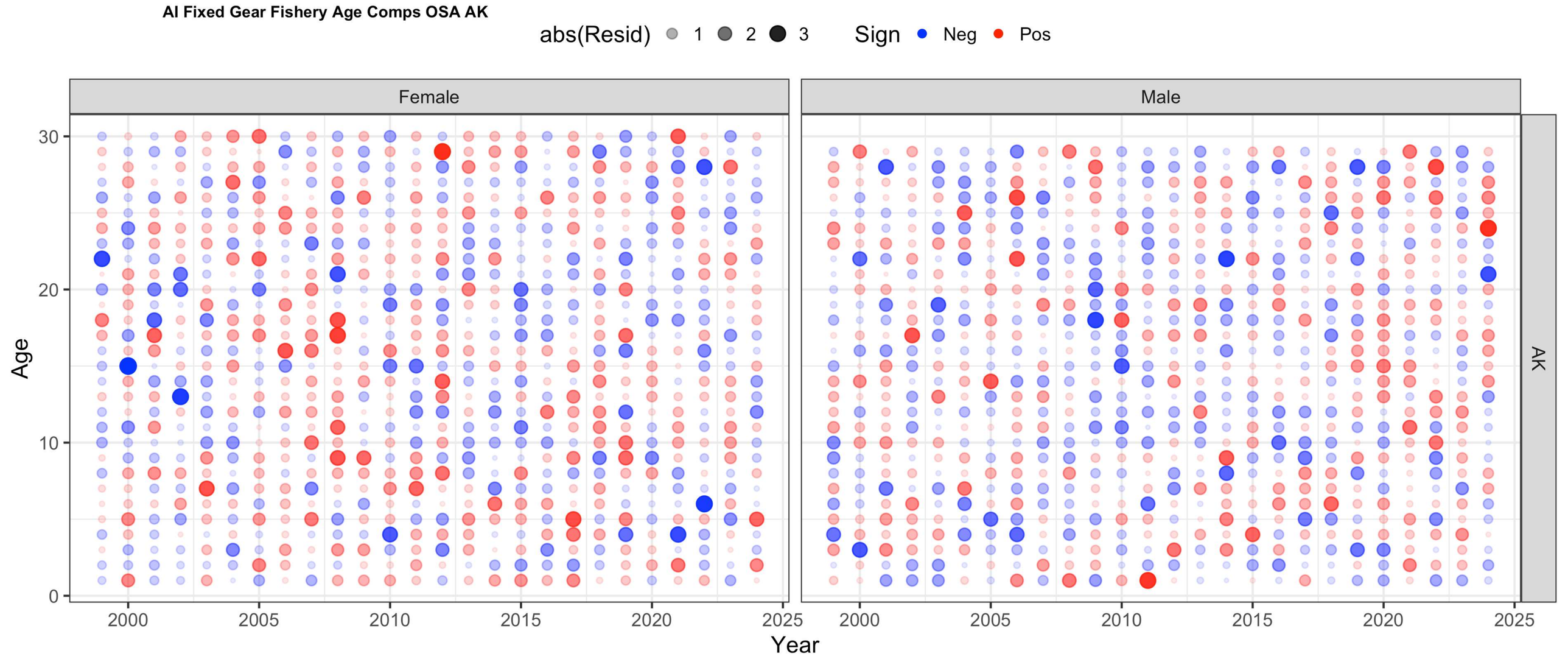
AI Fixed Gear Fishery Age Comps Female AK

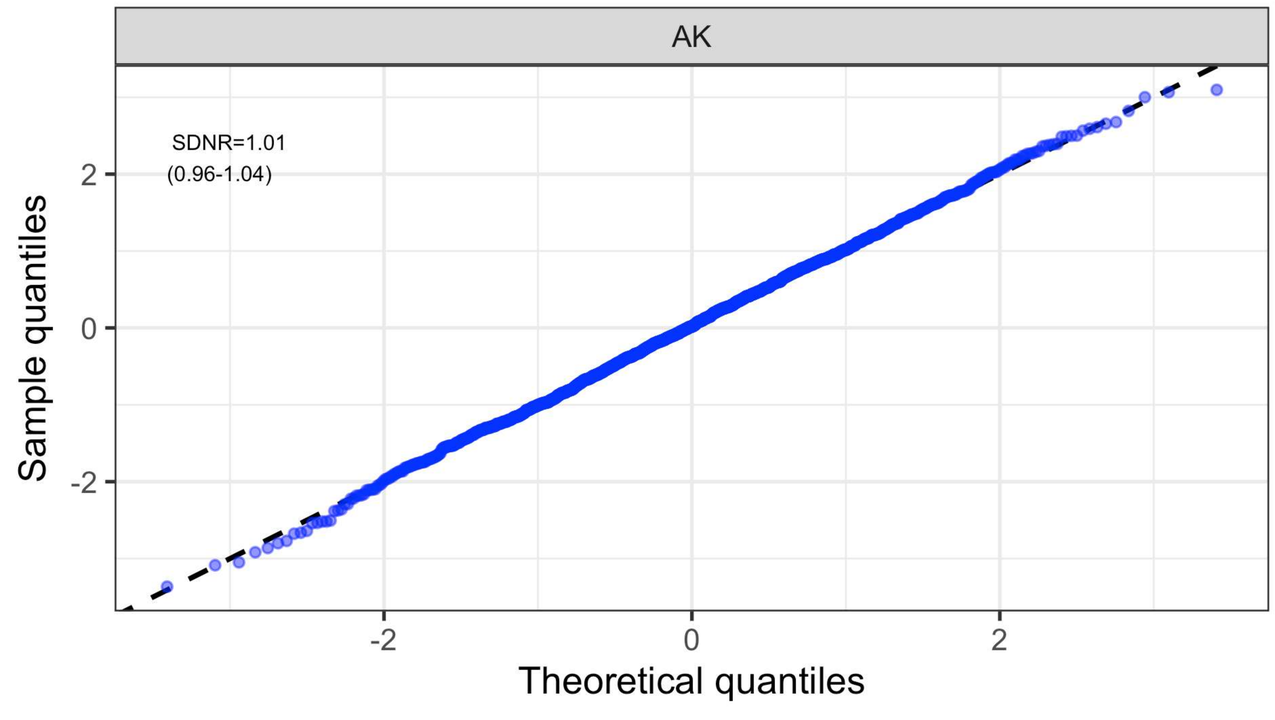


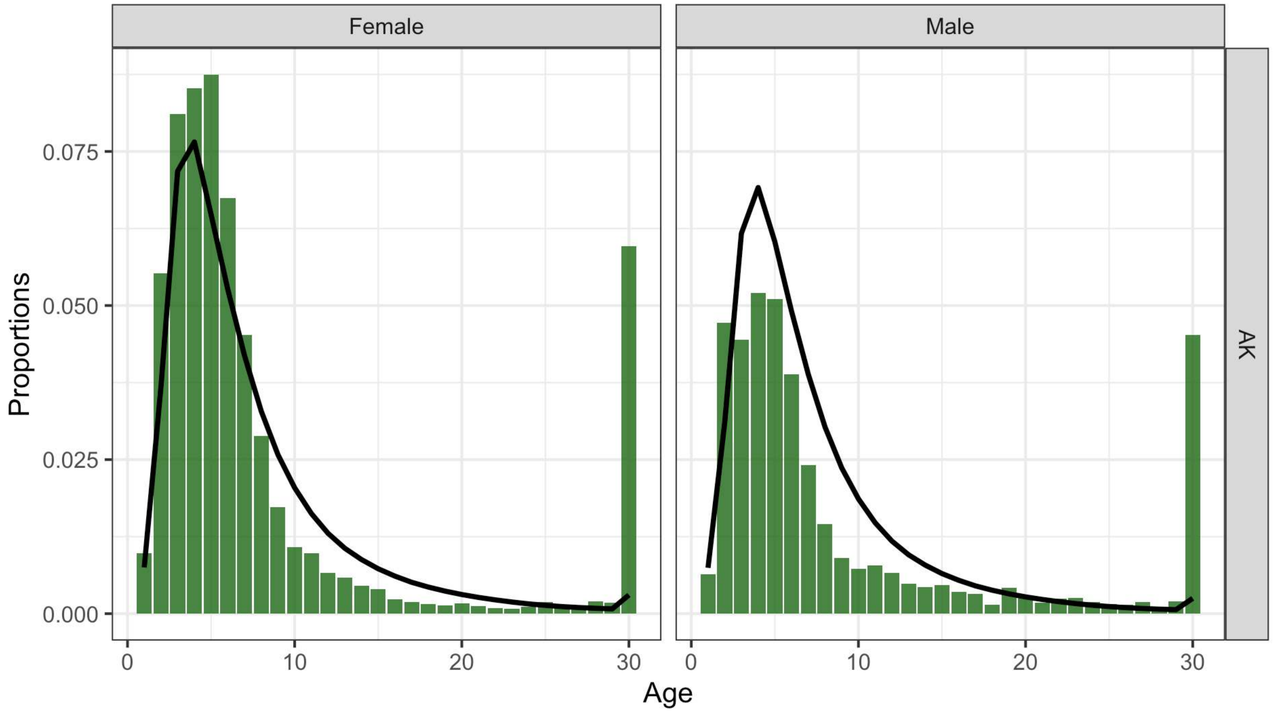
AI Fixed Gear Fishery Age Comps Male AK

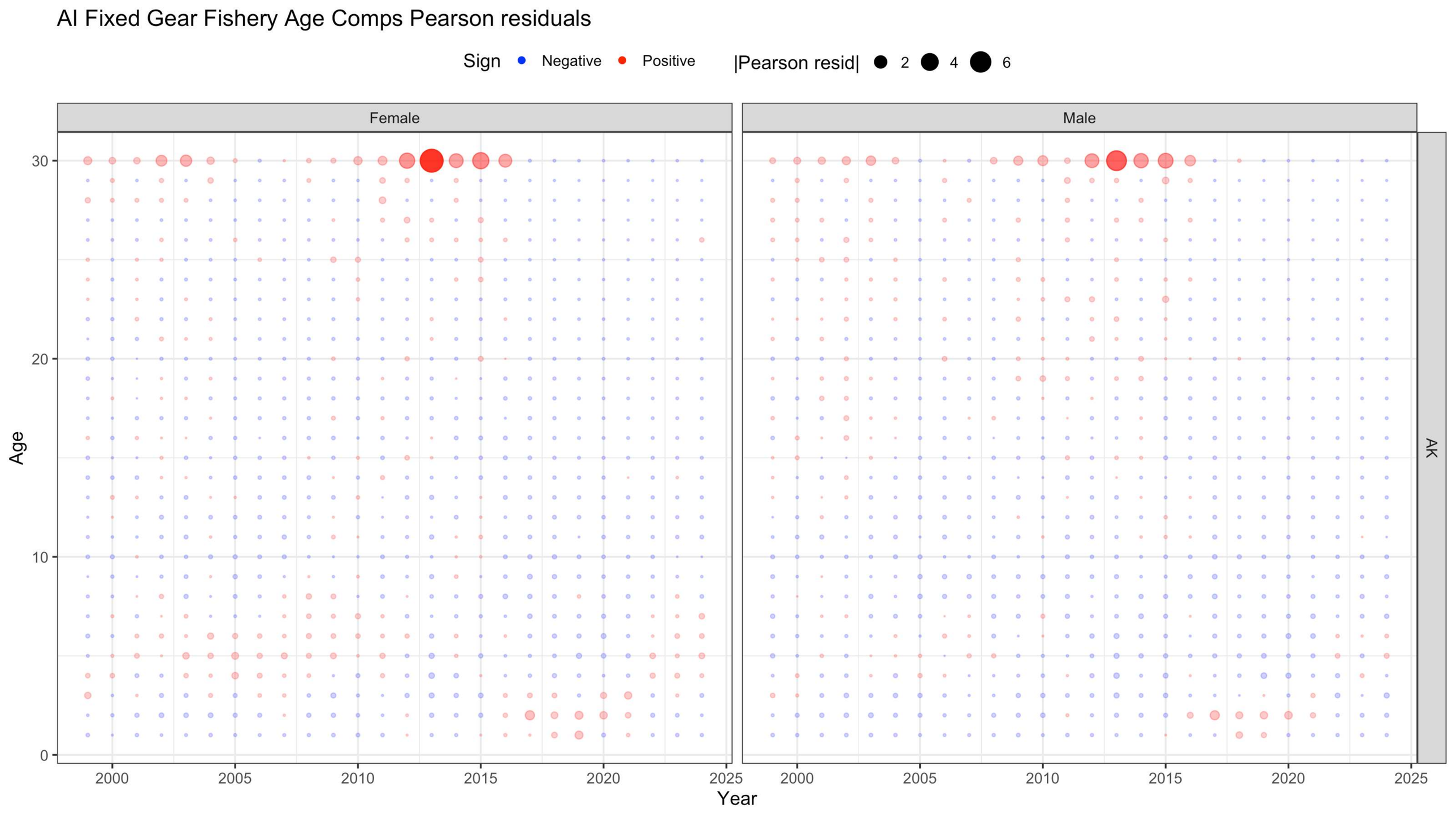


AI Fixed Gear Fishery Age Comps OSA AK








AI Fixed Gear Fishery Age Comps Pearson residuals
Sign
Negative
Positive
|Pearson resid|
2
4
6
Female
Male
AK
Age
Year
0
10
20
30
2000
2005
2010
2015
2020
2025

AI LL Survey Age Comps Female AK

1996
1998
2000
2002
2004
2006
2008
2010
2012
2014
2016
2018
2020
2022

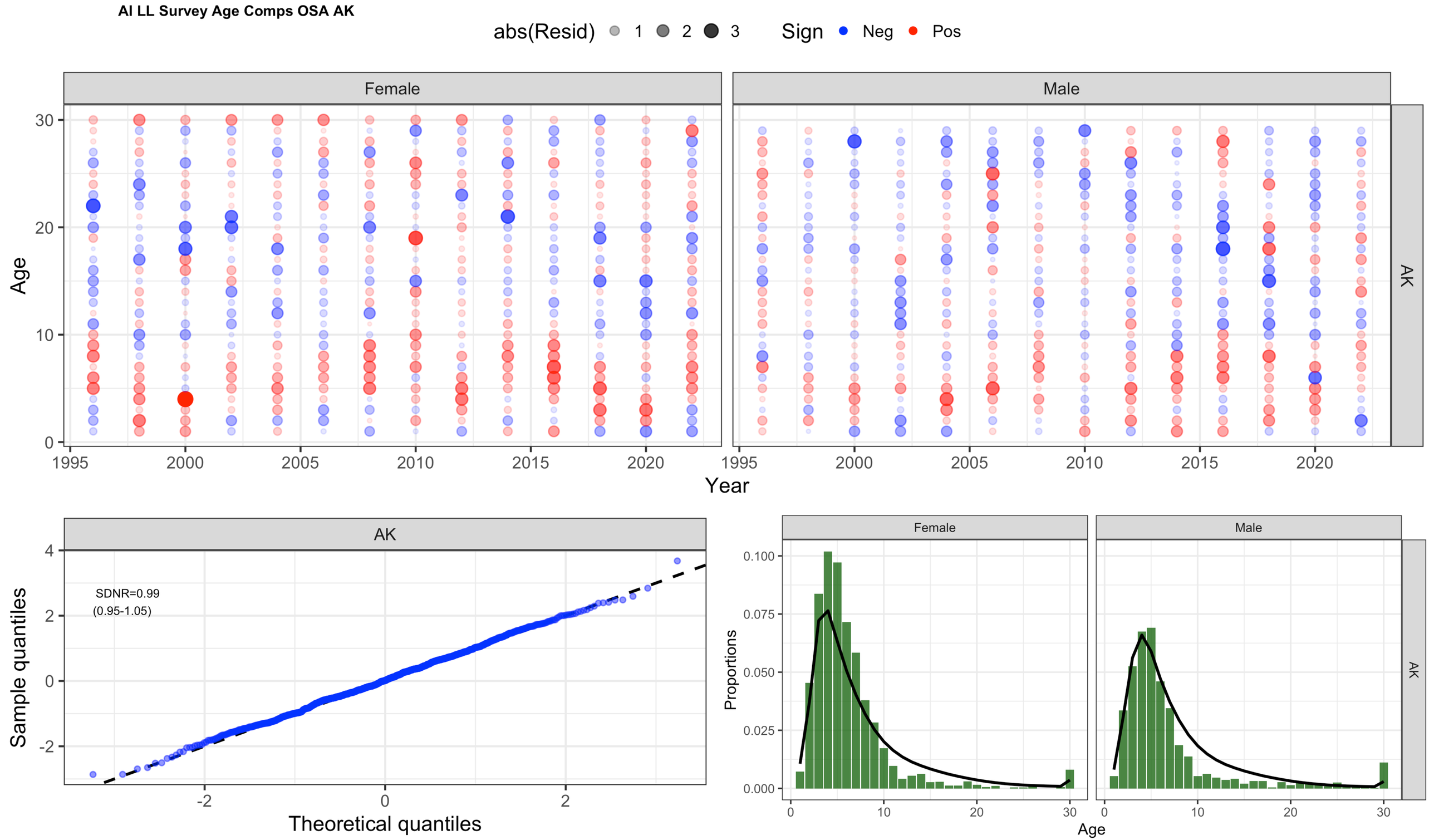
Proportion
0.00
0.05
0.10
0.15
0.20
0
10
20
30
Age

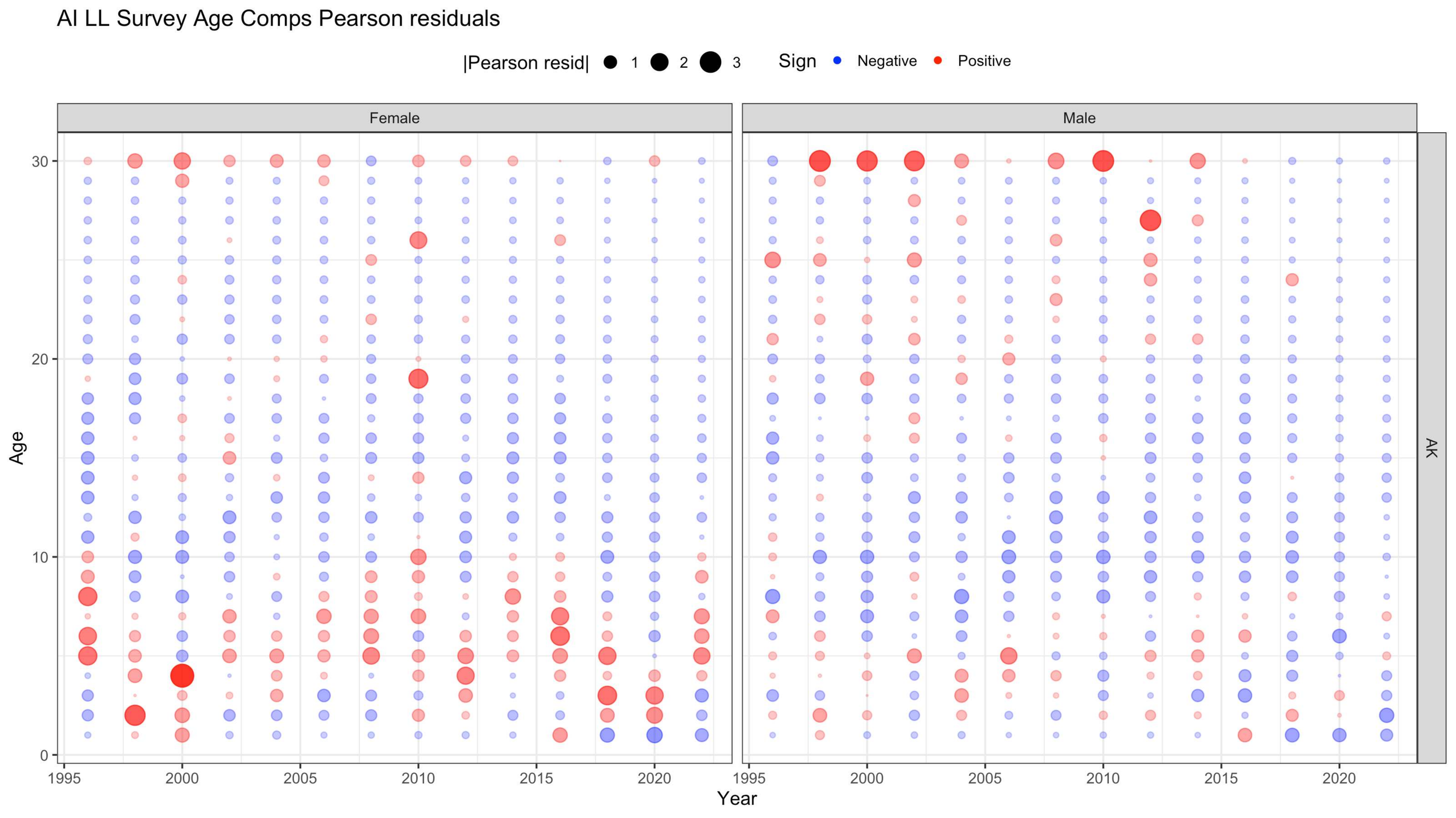
AI LL Survey Age Comps Male AK

1996 1998 2000 2002
2004 2006 2008 2010
2012 2014 2016 2018
2020 2022

Proportion
0.15 0.10 0.05 0.00
0 10 20 30
Age

AI LL Survey Age Comps OSA AK




AI LL Survey Age Comps Pearson residuals
|Pearson resid| 1 2 3
Sign Negative Positive
Female
Male
AK
Age
Year
0
10
20
30
1995
2000
2005
2010
2015
2020

Model 26.2_FAA_3LLF_LLS_BS_Dome_Gam Population Population 1

Female | Male | AK

Spawning Weight-at-Age (kg)

0.100
0.075
0.050
0.025
0.000

0 10 20 30

Age

Model 26.2_FAA_3LLF_LLS_BS_Dome_Gam Population Population 1

Female

Male

AK

Maturity at Age

1.00

0.75

0.50

0.25

0.00

0

10

20

30

Age

Model 26.2_FAA_3LLF_LLS_BS_Dome_Gam Population Population 1

Female

Male

AK

Spawning Weight at Age

6

4

2

0

0

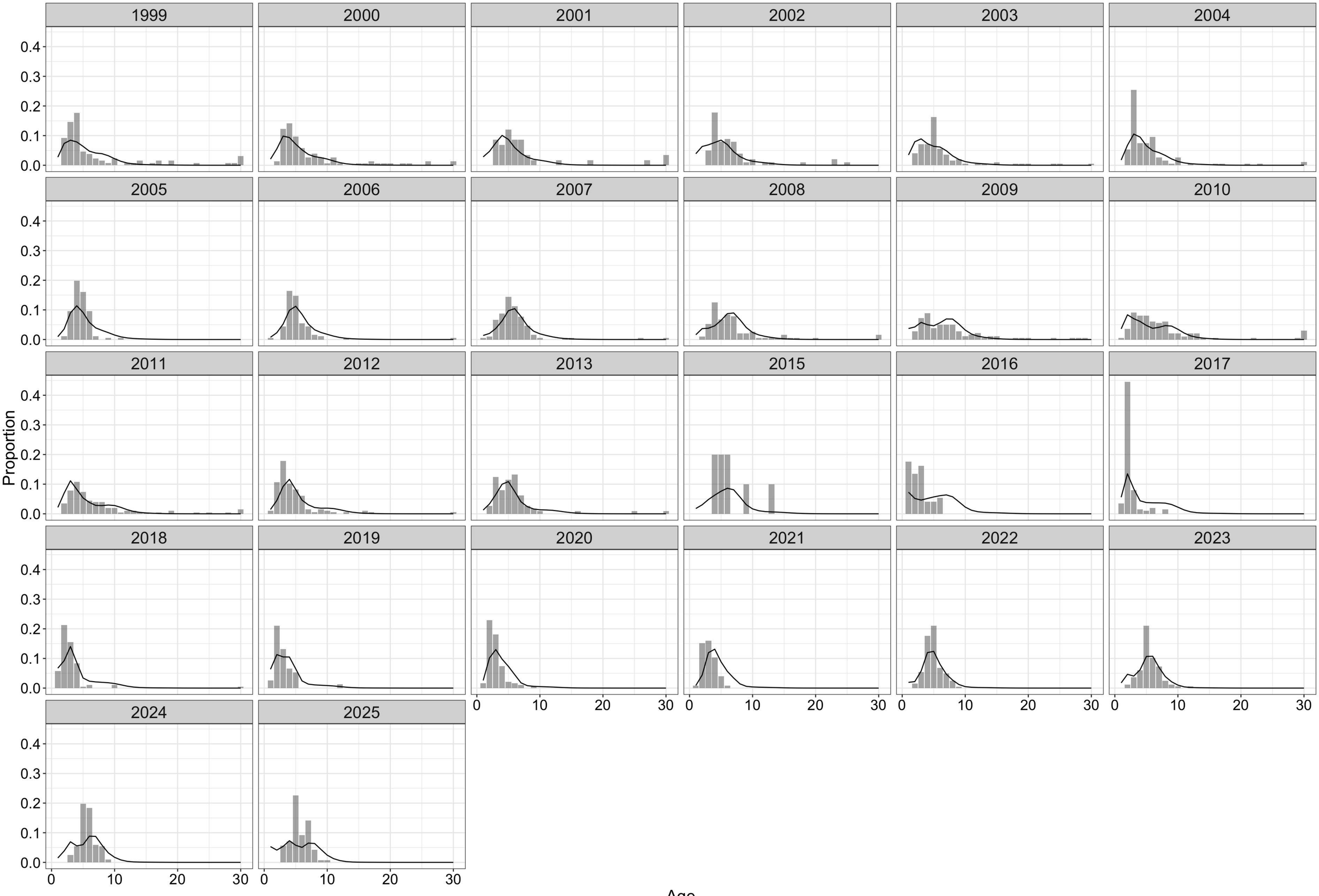
10

20

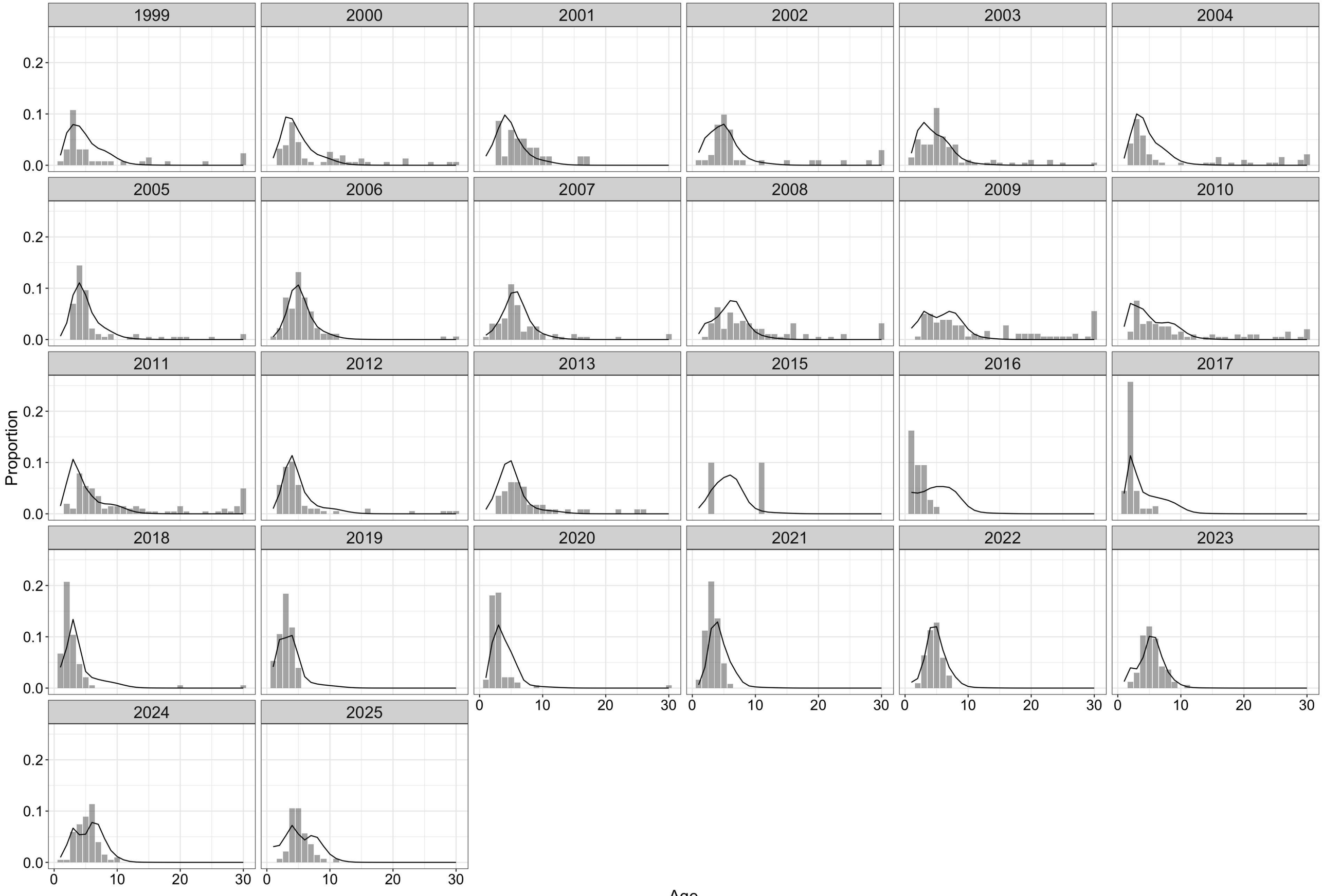
30

Age

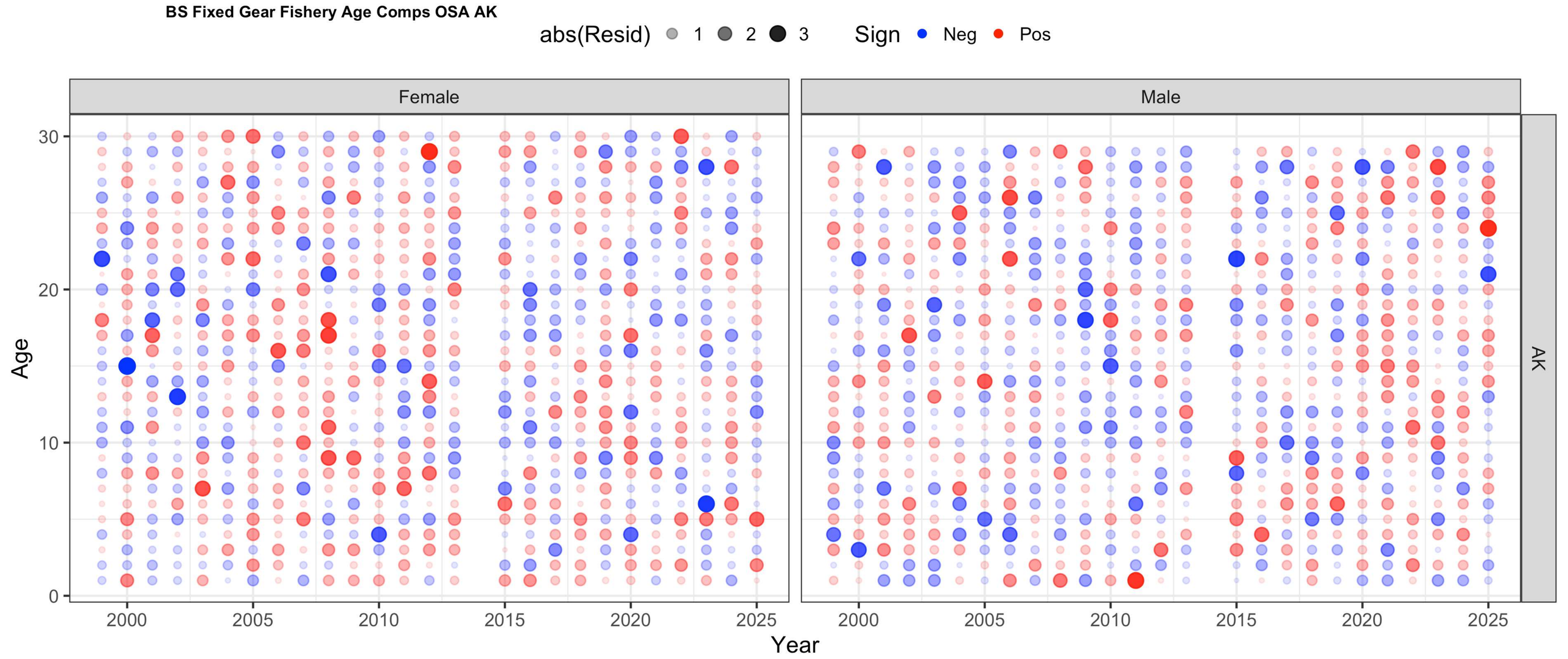
BS Fixed Gear Fishery Age Comps Female AK

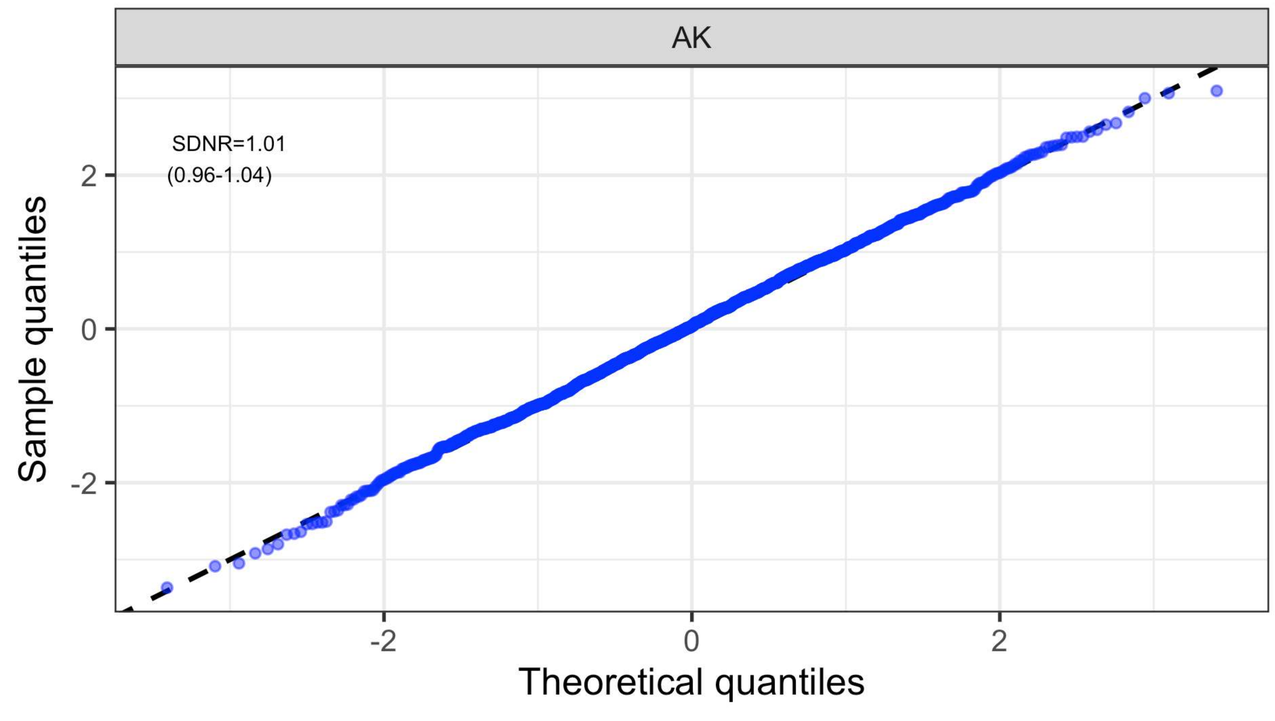


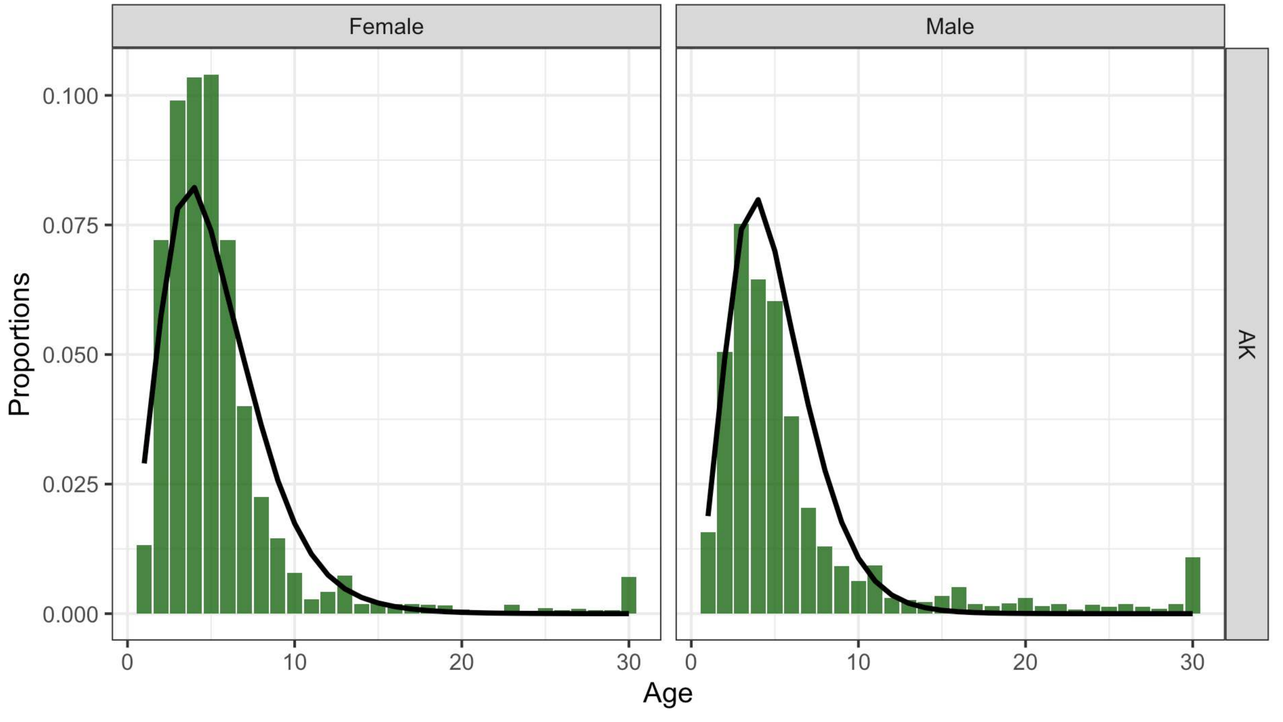
BS Fixed Gear Fishery Age Comps Male AK

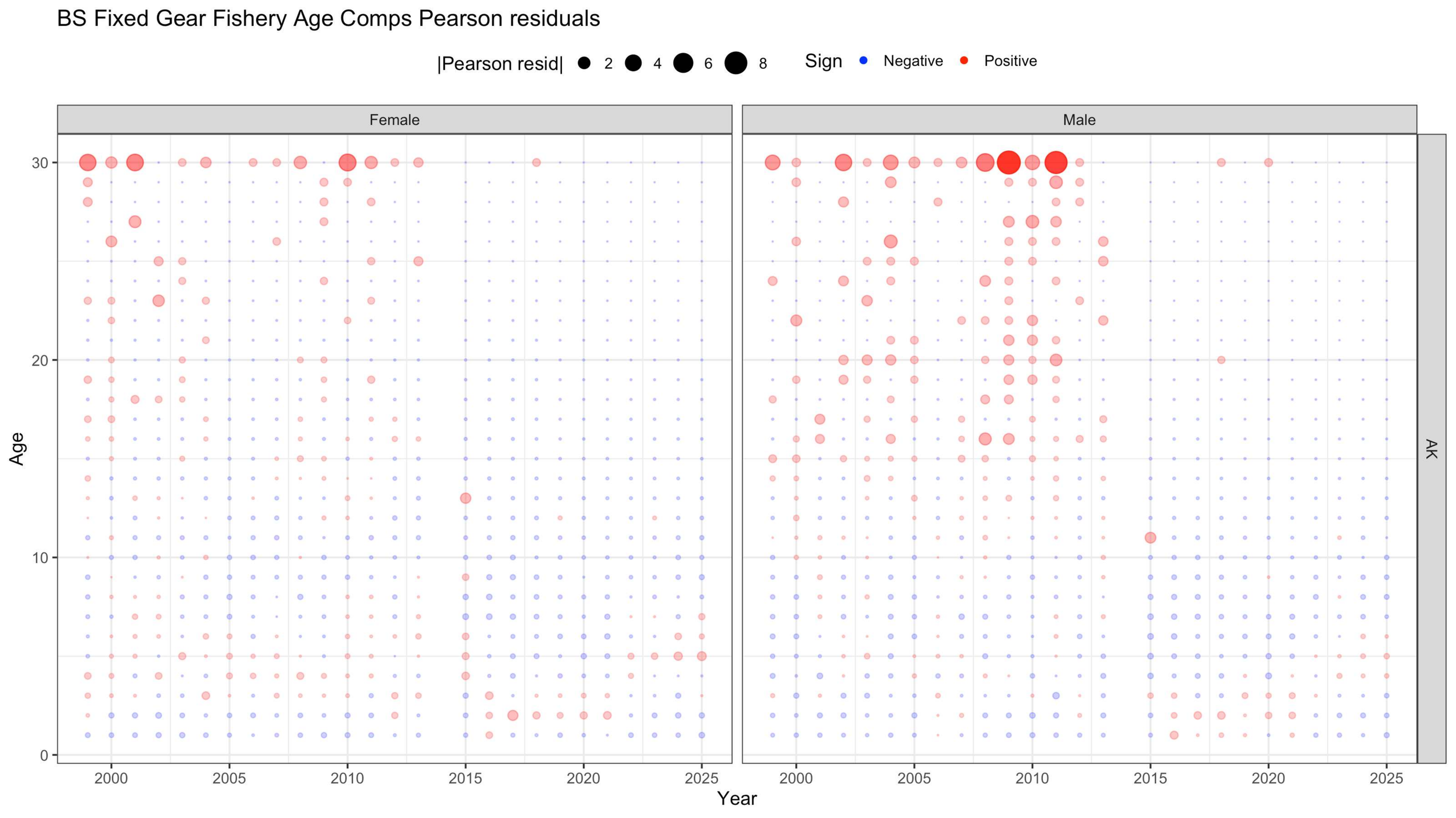


BS Fixed Gear Fishery Age Comps OSA AK








BS Fixed Gear Fishery Age Comps Pearson residuals
|Pearson resid|
2
4
6
8
Sign
Negative
Positive
Female
Male
AK
Age
Year
0
10
20
30
2000
2005
2010
2015
2020
2025

BS LL Survey Age Comps Female AK

1997 1999 2001 2003
2005 2007 2009 2011
2013 2015 2017 2019
2021 2023

Proportion

0.0 0.1 0.2 0.3

0 10 20 30

Age

BS LL Survey Age Comps Male AK

1997 1999 2001 2003
2005 2007 2009 2011
2013 2015 2017 2019
2021 2023

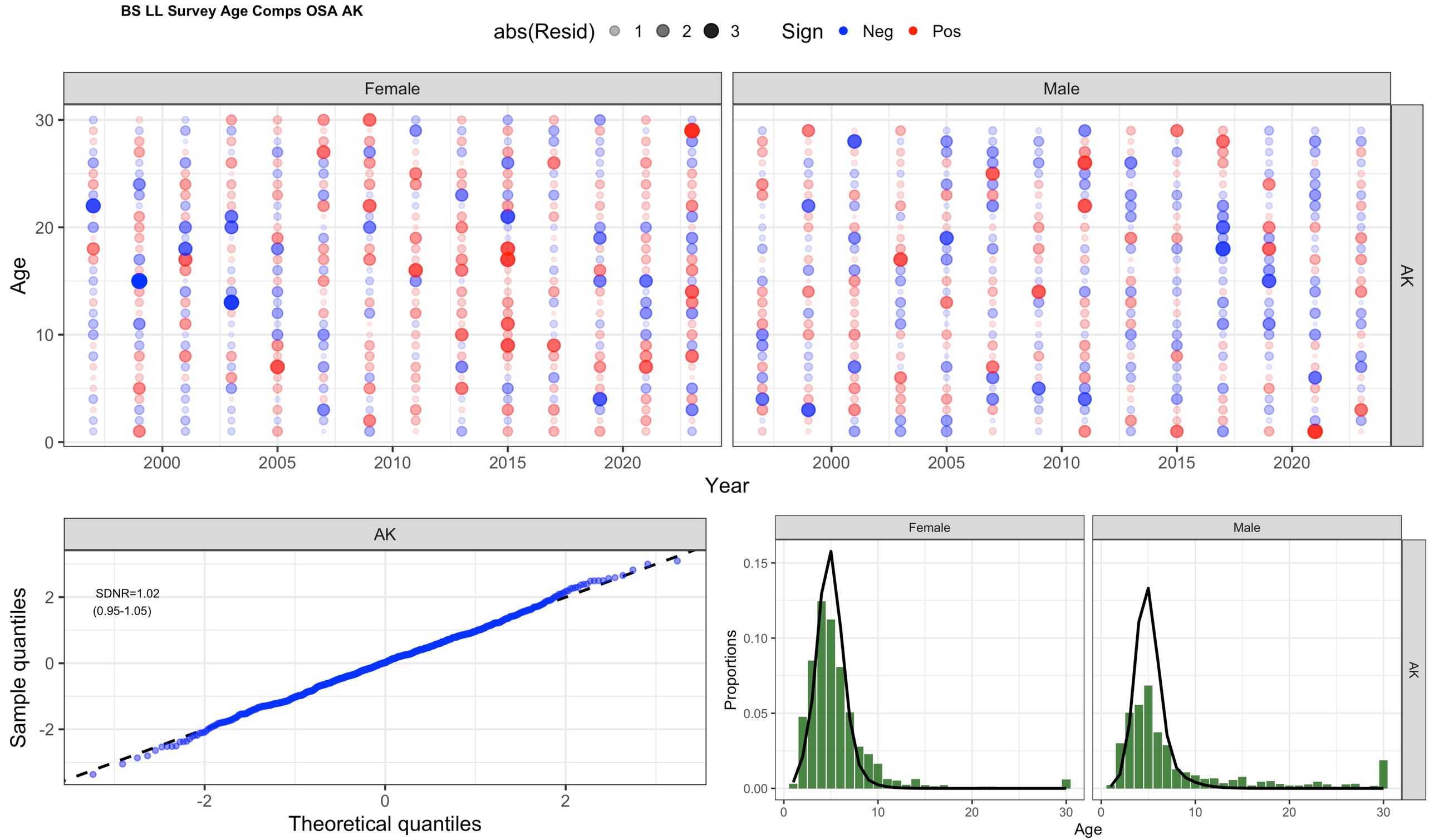
Proportion

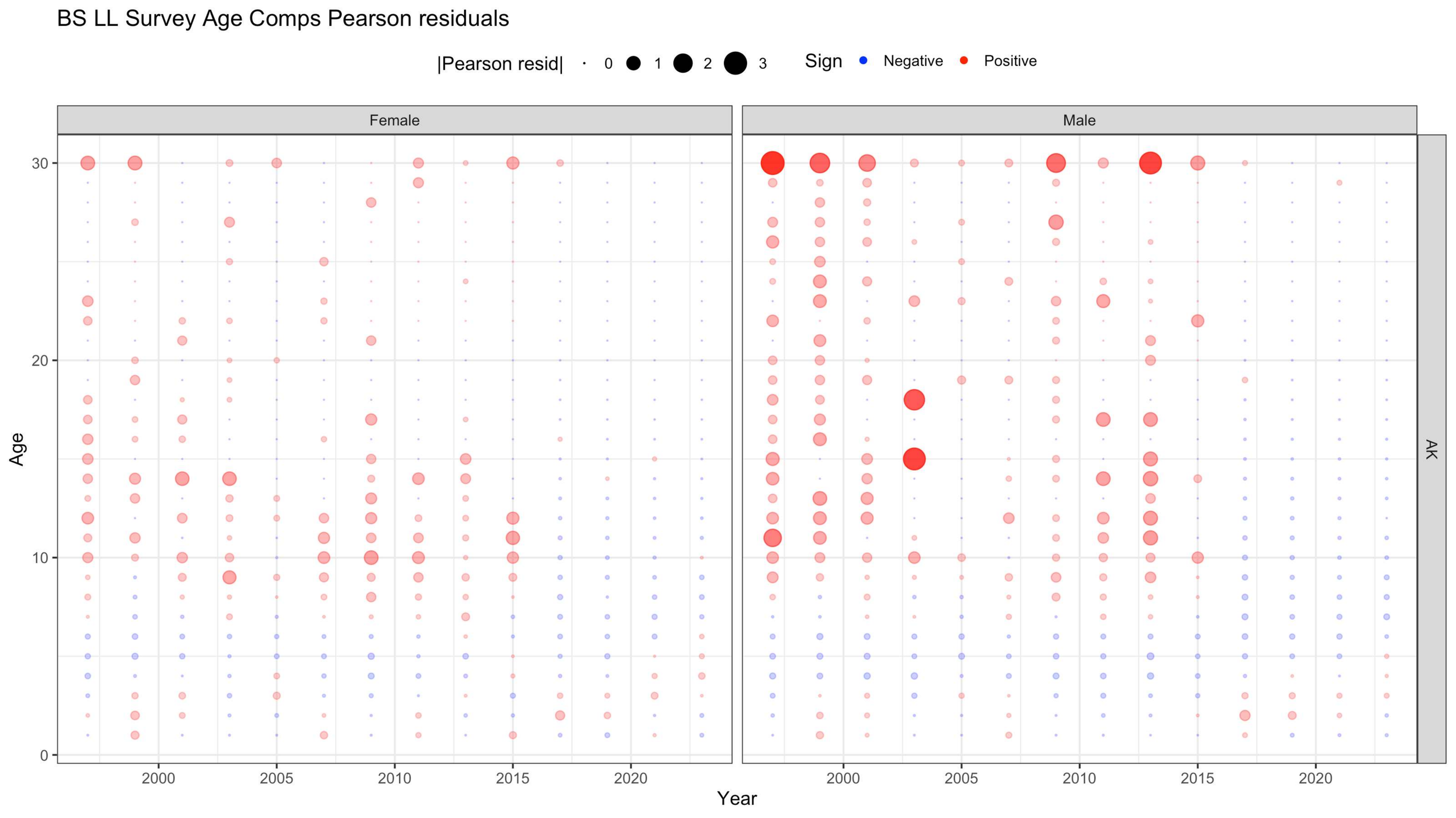
0.00 0.05 0.10 0.15

0 10 20 30

Age

BS LL Survey Age Comps OSA AK




BS LL Survey Age Comps Pearson residuals
|Pearson resid|
0
1
2
3
Sign
Negative
Positive
Female
Male
AK
Age
Year
0
10
20
30
2000
2005
2010
2015
2020

Model — 26.2_FAA_3LLF_LLS_BS_Dome_Gam

AK

BS LLF

AI LLF

GOA LLF

TF

Catch

Year

0
5
10
15
0
1
2
3
0
10
20
30
40
0
10
20
1960
1980
2000
2020

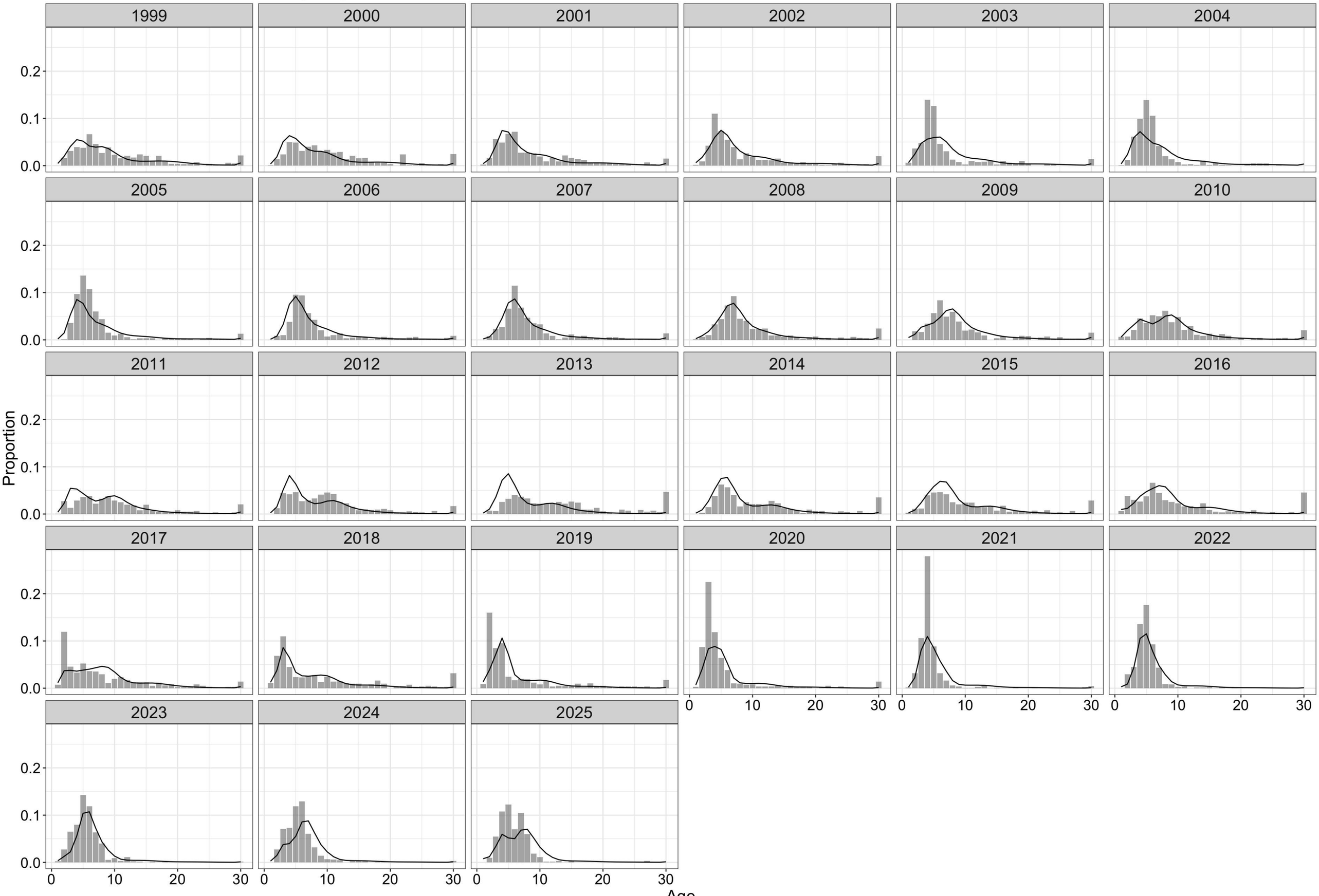
AK

JPN LLS Lengths
JPN LLS Index
GOA LLS Index
AI LLS Index
BS LLS Index
JPN LLS Ages
GOA LLS Ages
AI LLS Ages
BS LLS Ages
TF Lengths
TF Catch
GOA LLF Catch
AI LLF Catch
BS LLF Catch
GOA LLF Ages
AI LLF Ages
BS LLF Ages

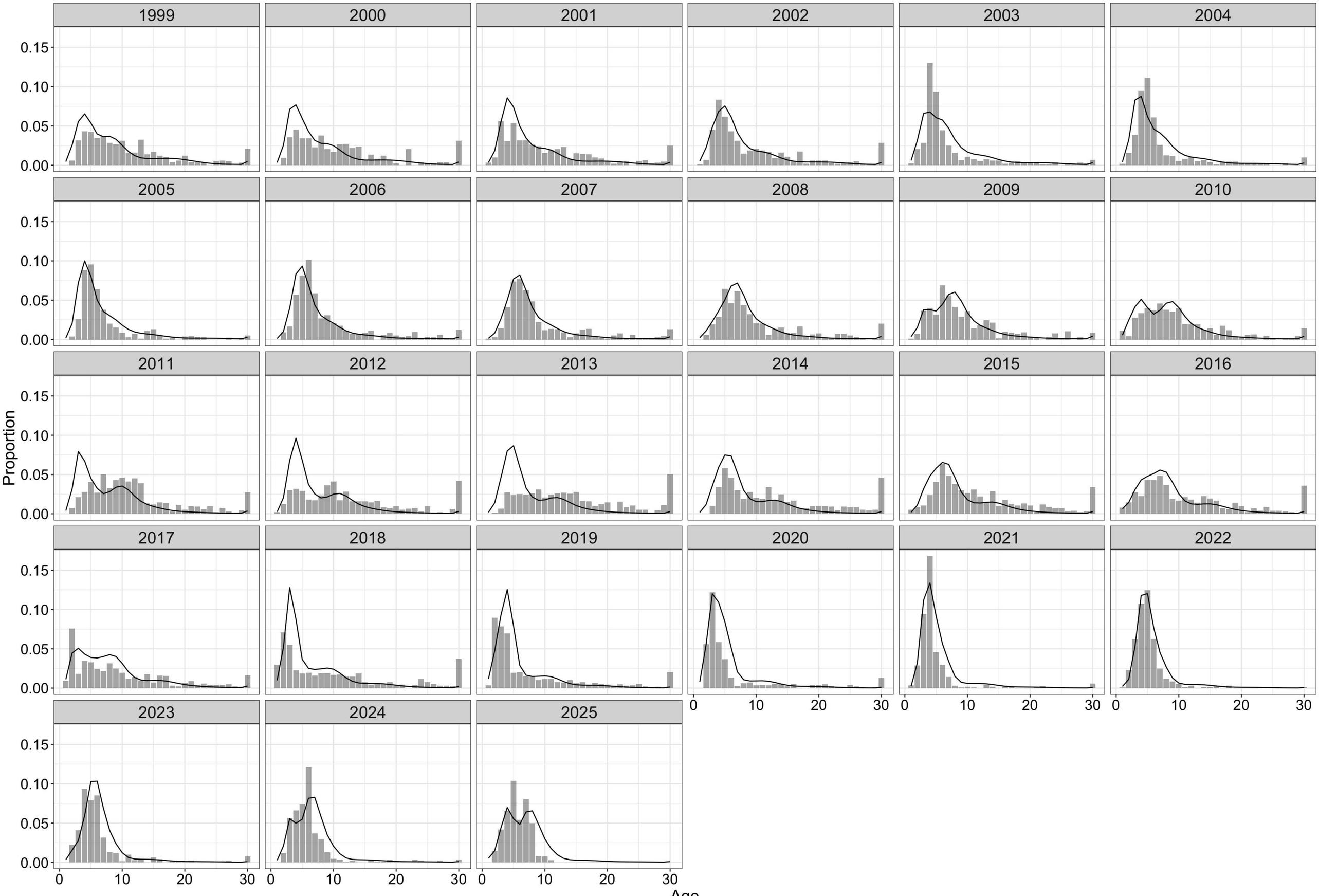
1960
1980
2000
2020
Year

26.2_FAA_3LLF_LLS_BS_Dome_Gam

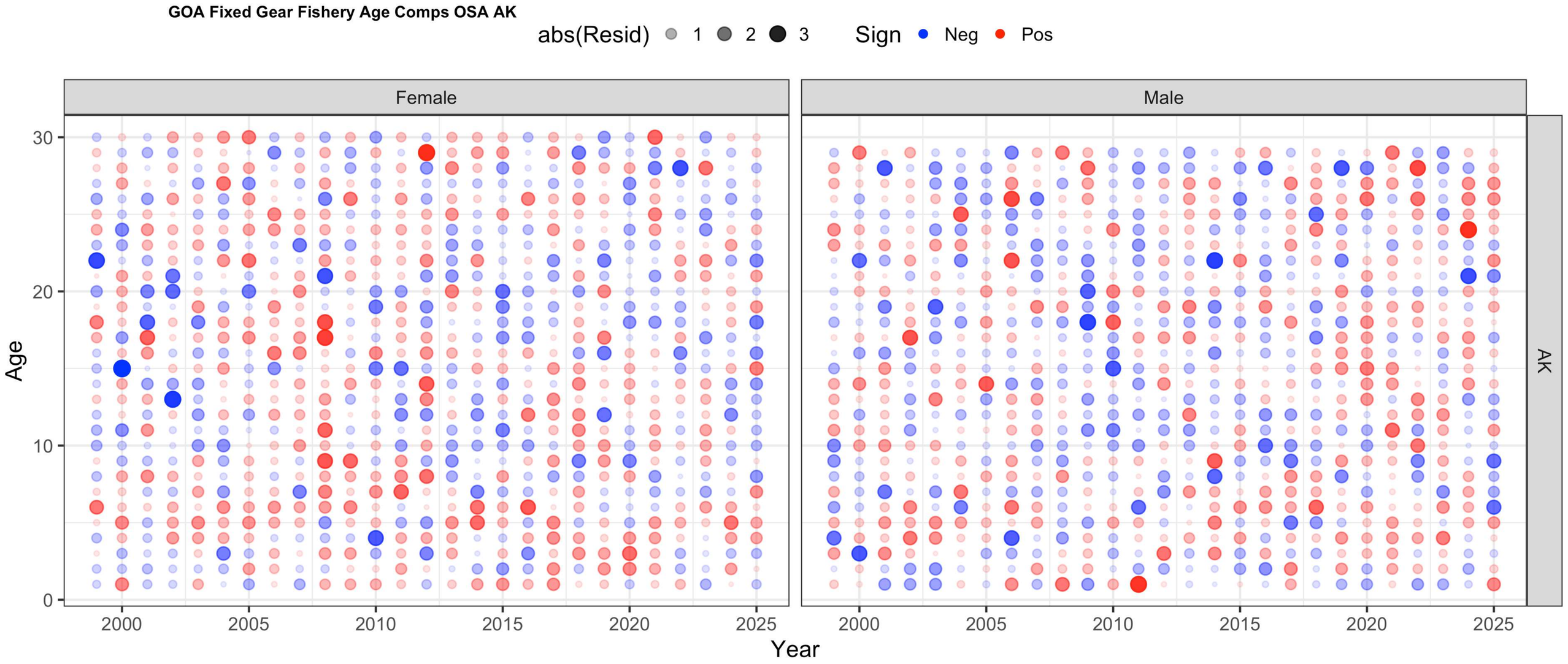
GOA Fixed Gear Fishery Age Comps Female AK

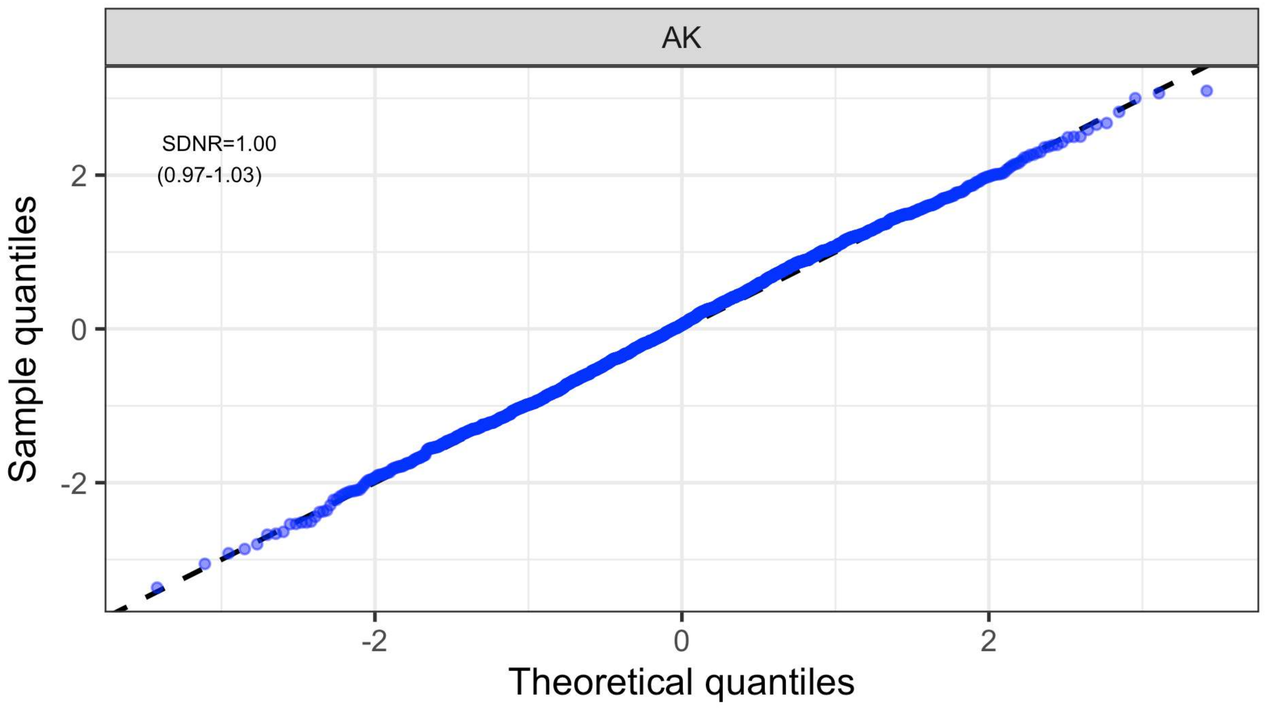


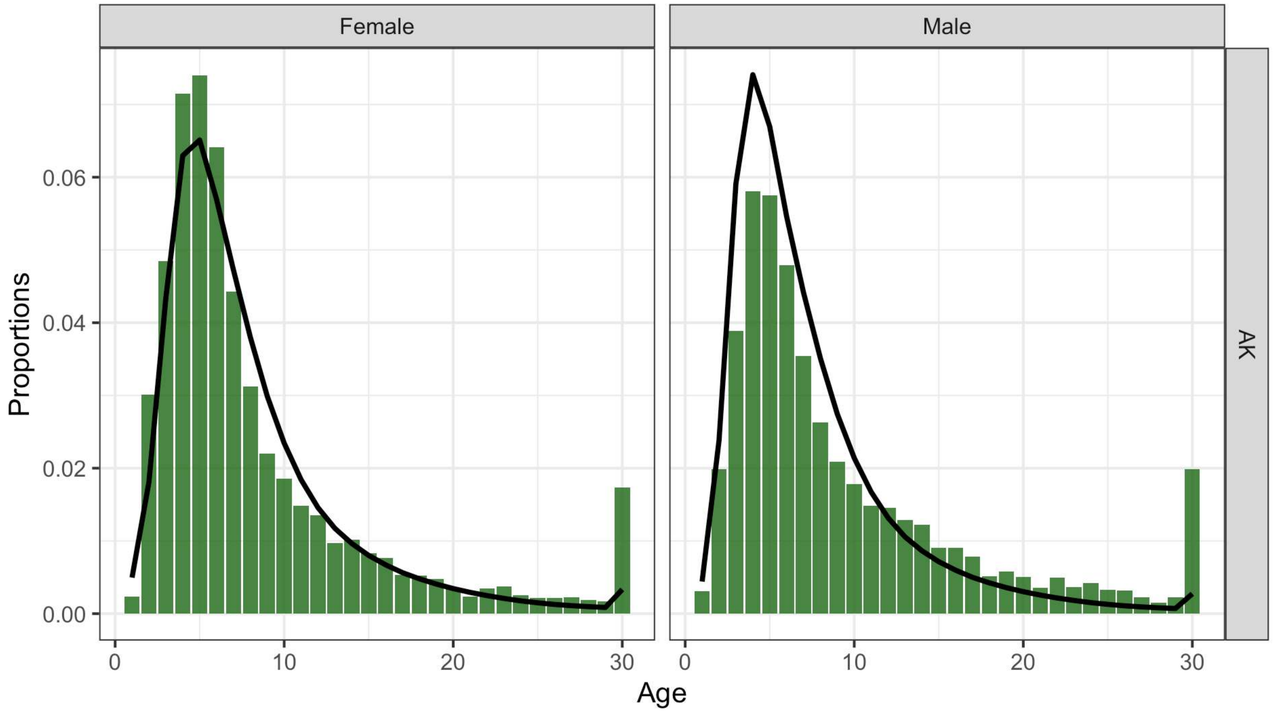
GOA Fixed Gear Fishery Age Comps Male AK

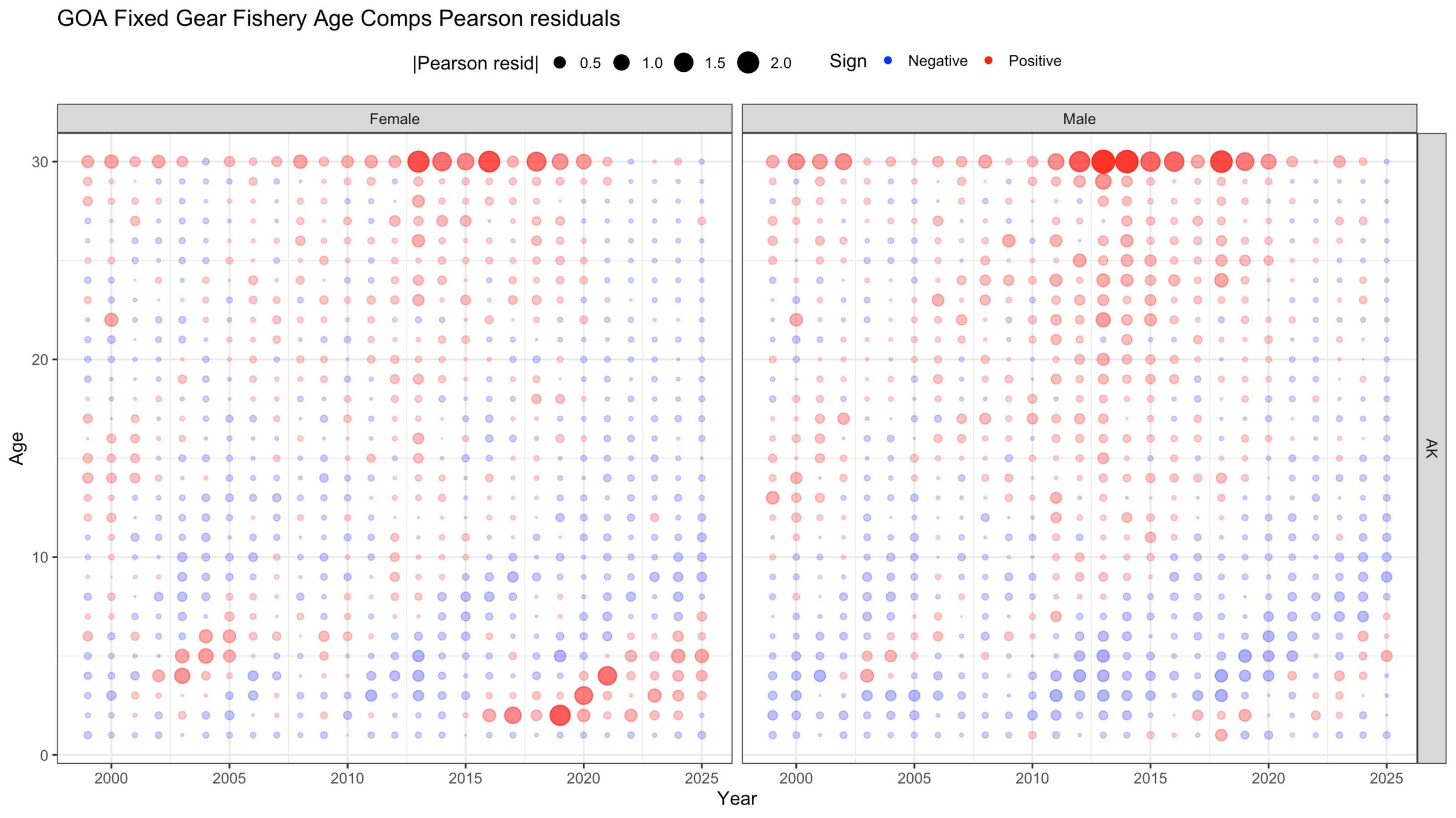


GOA Fixed Gear Fishery Age Comps OSA AK








GOA Fixed Gear Fishery Age Comps Pearson residuals
|Pearson resid|
0.5
1.0
1.5
2.0
Sign
Negative
Positive
Female
Male
AK
Age
Year
0
10
20
30
2000
2005
2010
2015
2020
2025

GOA LL Survey Age Comps Female AK

1996 1997 1998 1999 2000 2001
2002 2003 2004 2005 2006 2007
2008 2009 2010 2011 2012 2013
2014 2015 2016 2017 2018 2019
2020 2021 2022 2023 2025

Proportion
0.0 0.1 0.2
0 10 20 30
Age

GOA LL Survey Age Comps Male AK

1996 1997 1998 1999 2000 2001
2002 2003 2004 2005 2006 2007
2008 2009 2010 2011 2012 2013
2014 2015 2016 2017 2018 2019
2020 2021 2022 2023 2025

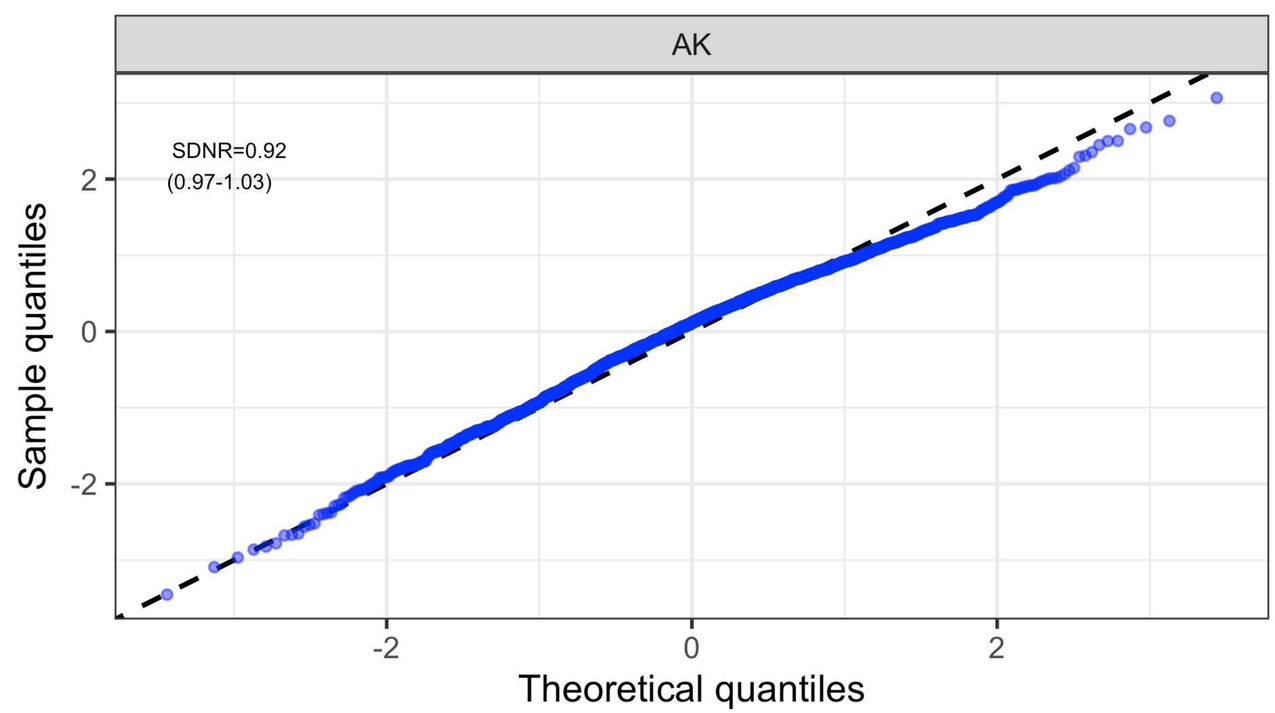
Proportion

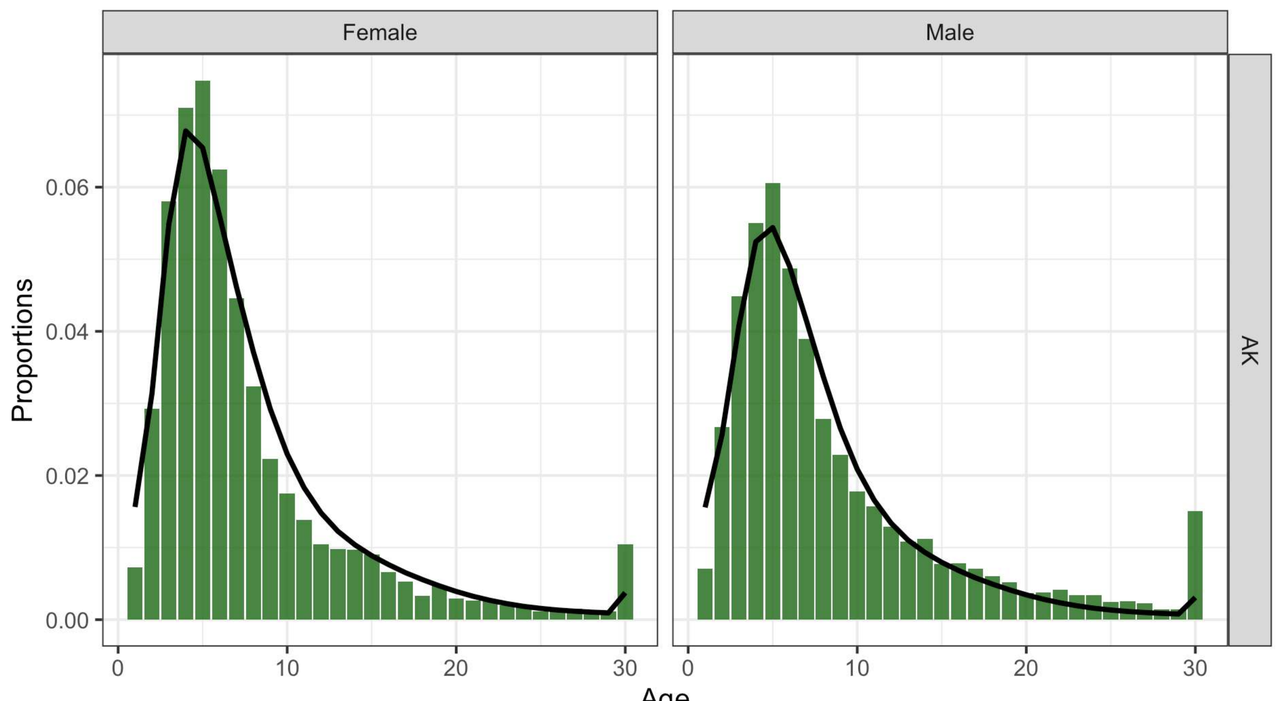
0.00 0.05 0.10 0.15

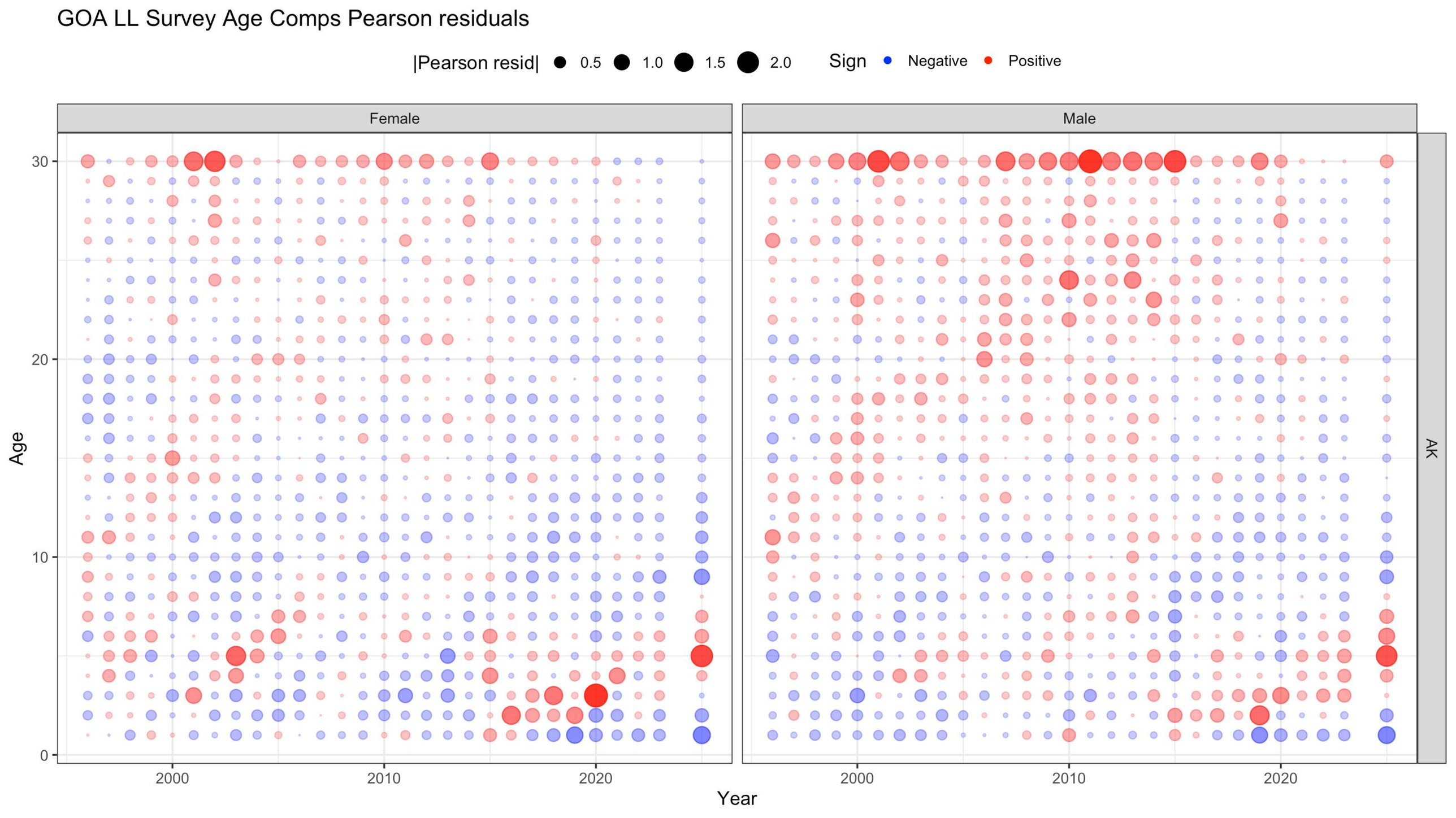
0 10 20 30

Age

GOA LL Survey Age Comps OSA AK






GOA LL Survey Age Comps Pearson residuals
|Pearson resid|
0.5
1.0
1.5
2.0
Sign
Negative
Positive
Female
Male
AK
Age
Year
0
10
20
30
2000
2010
2020

Model — 26.2_FAA_3LLF_LLS_BS_Dome_Gam

BS LLS

AI LLS

GOA LLS

JPN LLS

RPNs (1000s)

Year

Jitter SSB and Recruitment

Region 1

Recruitment (millions)

SSB (kt)

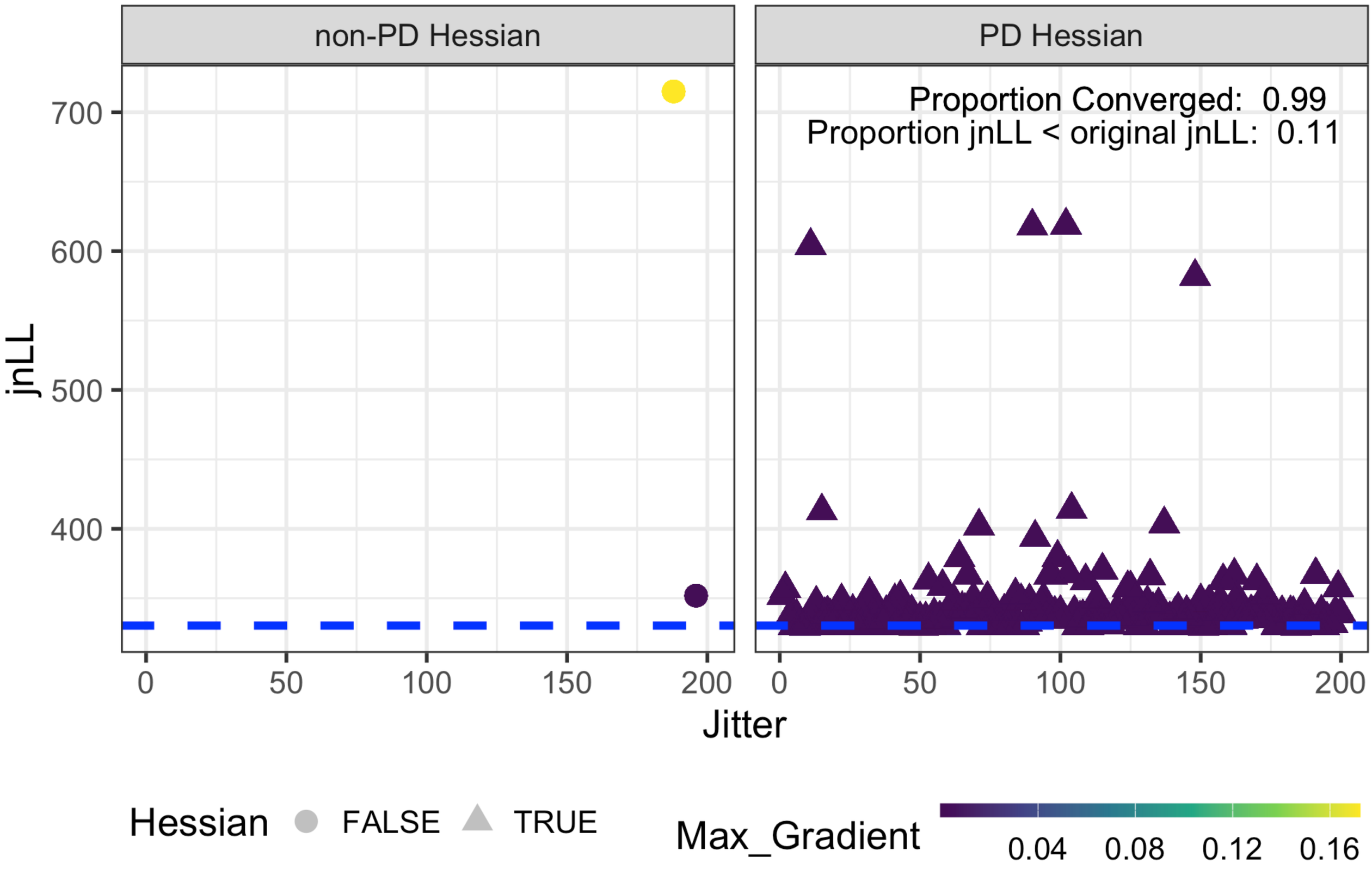
Proportion Converged: 0.99
Proportion jnLL < original jnLL: 0.11

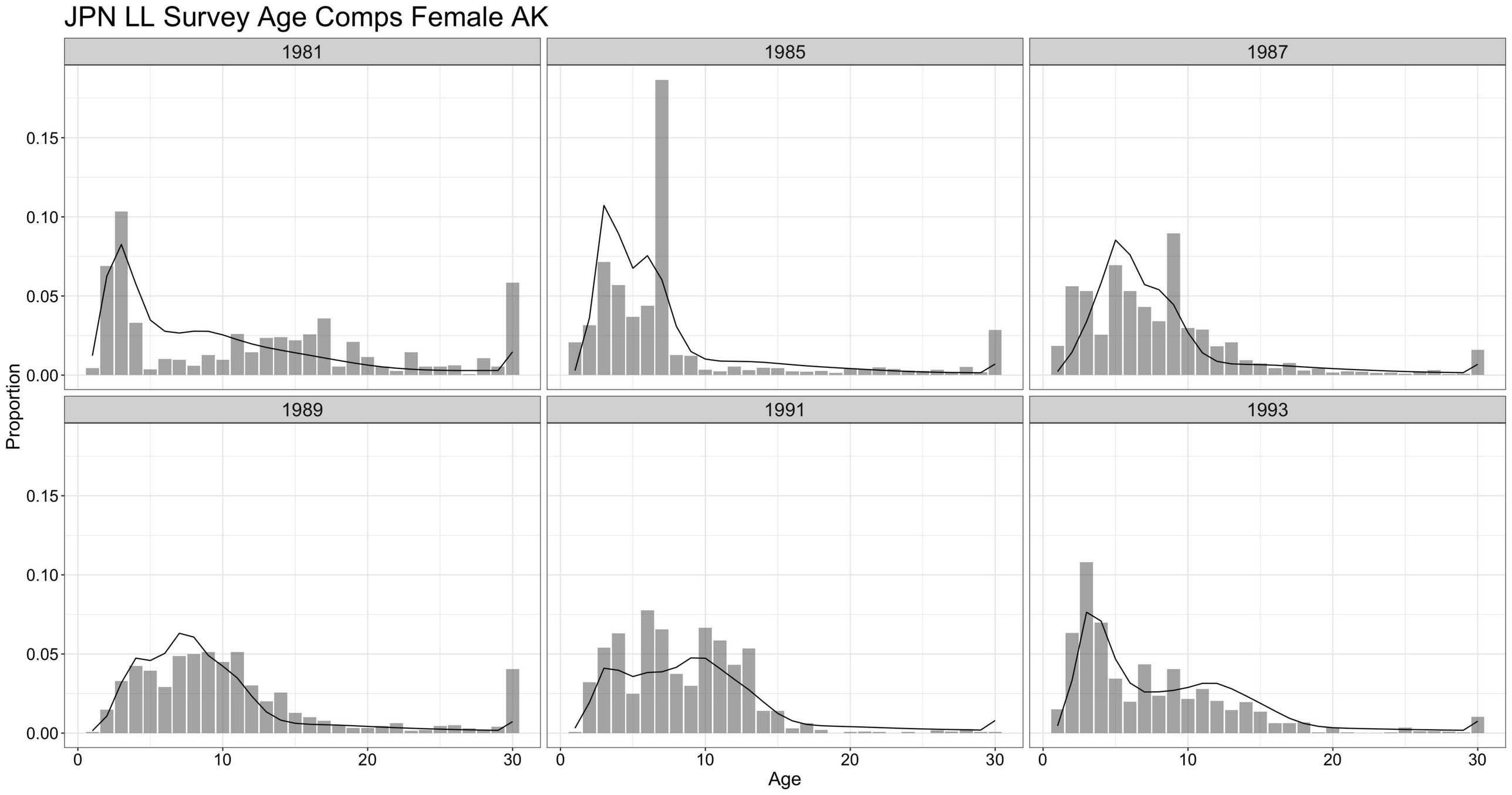
Hessian
FALSE
TRUE

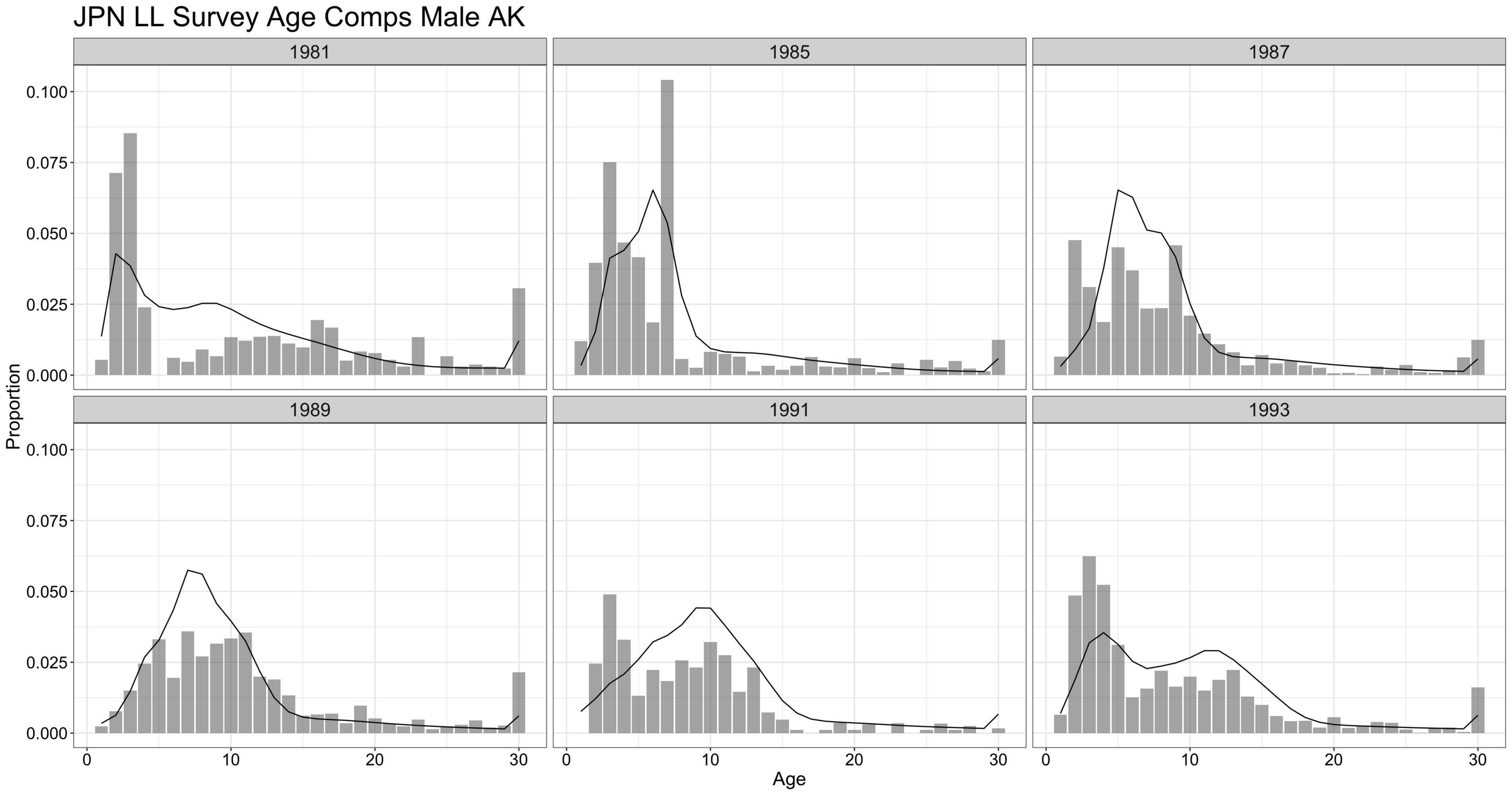
Value

Year

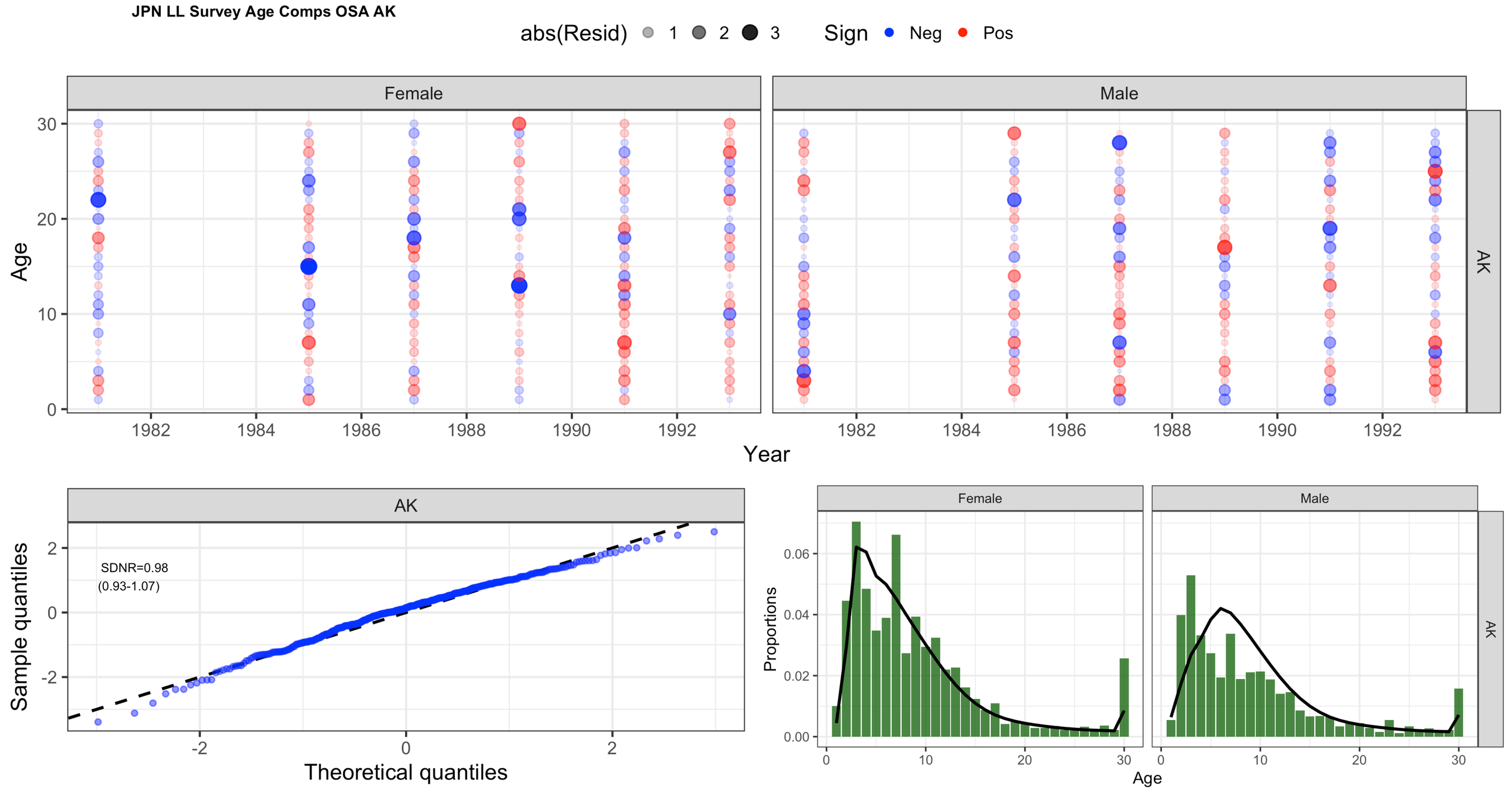
Jitter nLL and Convergence

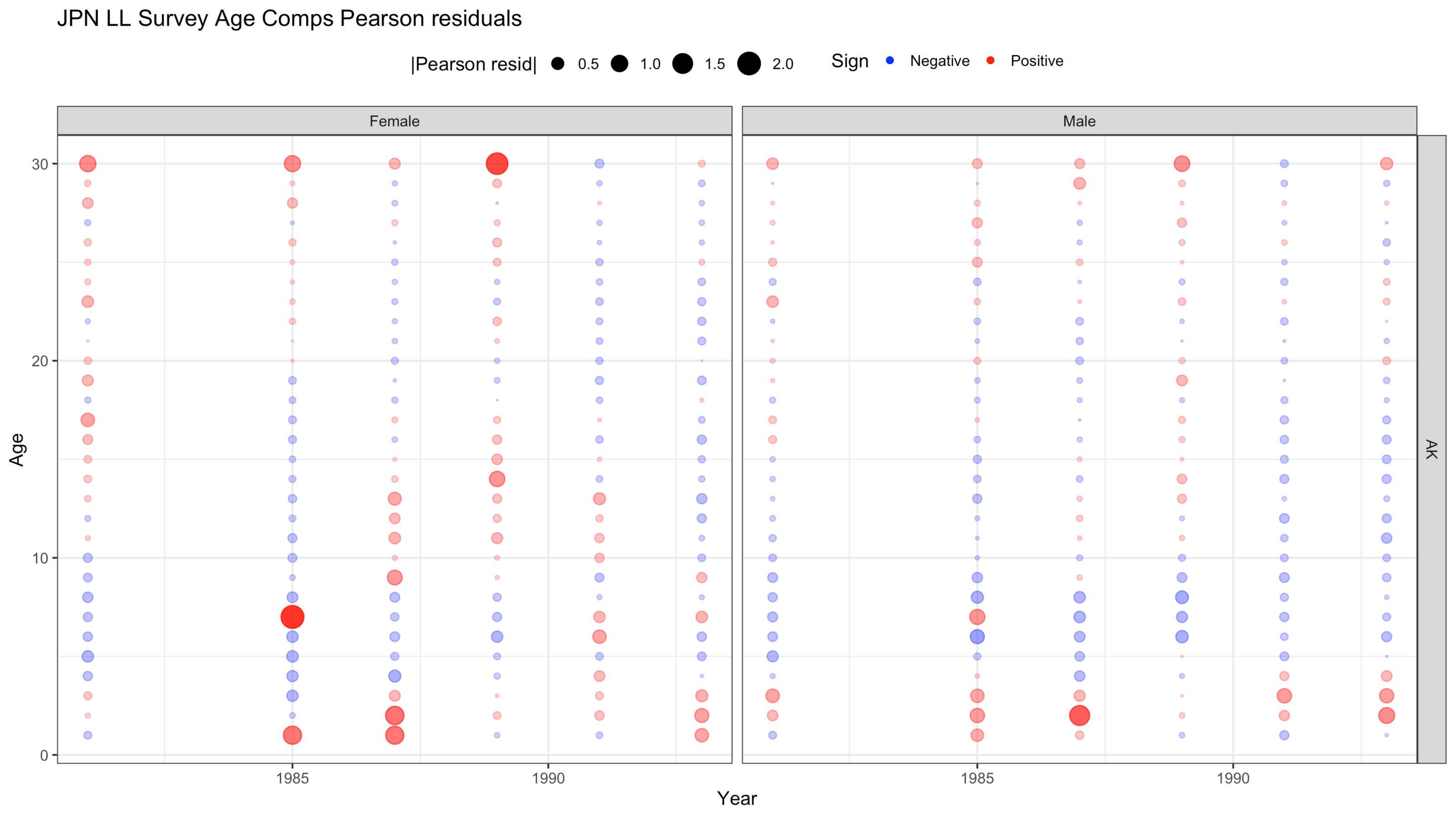



JPN LL Survey Age Comps Female AK
1981
1985
1987
1989
1991
1993
Proportion
0.00
0.05
0.10
0.15
0
10
20
30
Age


JPN LL Survey Age Comps Male AK
1981
1985
1987
1989
1991
1993
0.100
0.075
0.050
0.025
0.000
Proportion
0
10
20
30
Age

JPN LL Survey Age Comps OSA AK




JPN LL Survey Age Comps Pearson residuals
|Pearson resid|
0.5
1.0
1.5
2.0
Sign
Negative
Positive
Female
Male
AK
Age
30
20
10
0
1985
1990
1985
1990
Year

JPN LL Survey Length Comps Female AK

1979 1980 1981 1982

1983 1984 1985 1986

1987 1988 1989 1990

1991 1992 1993 1994

Proportion

0.00 0.02 0.04 0.06 0.08

40 60 80 100

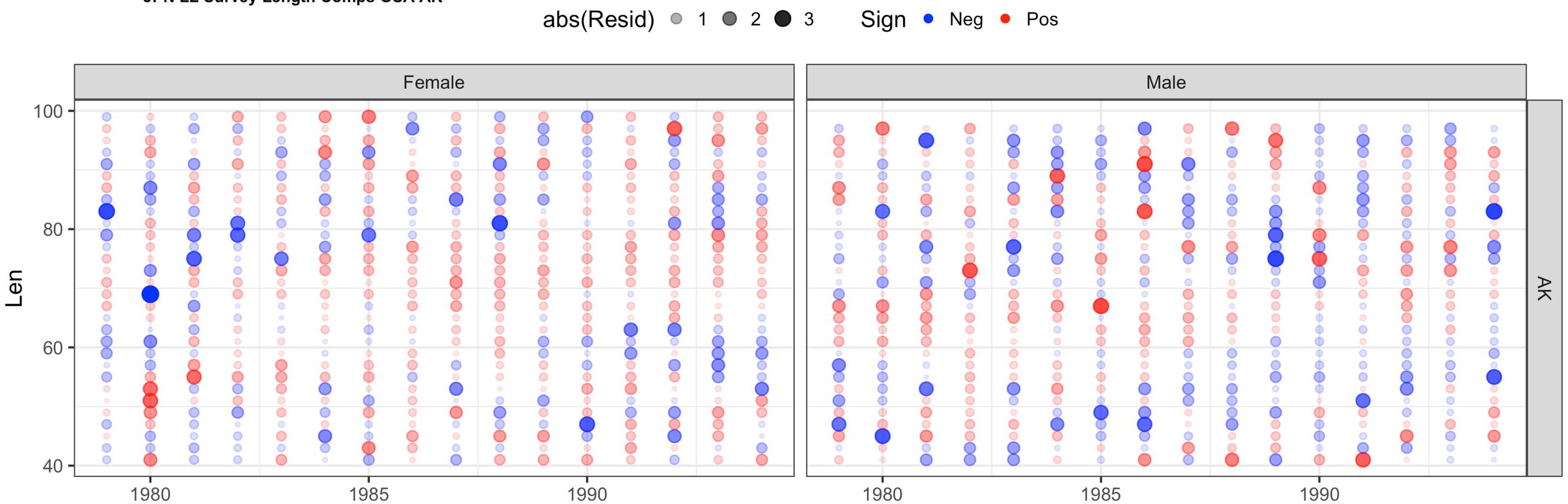
Len

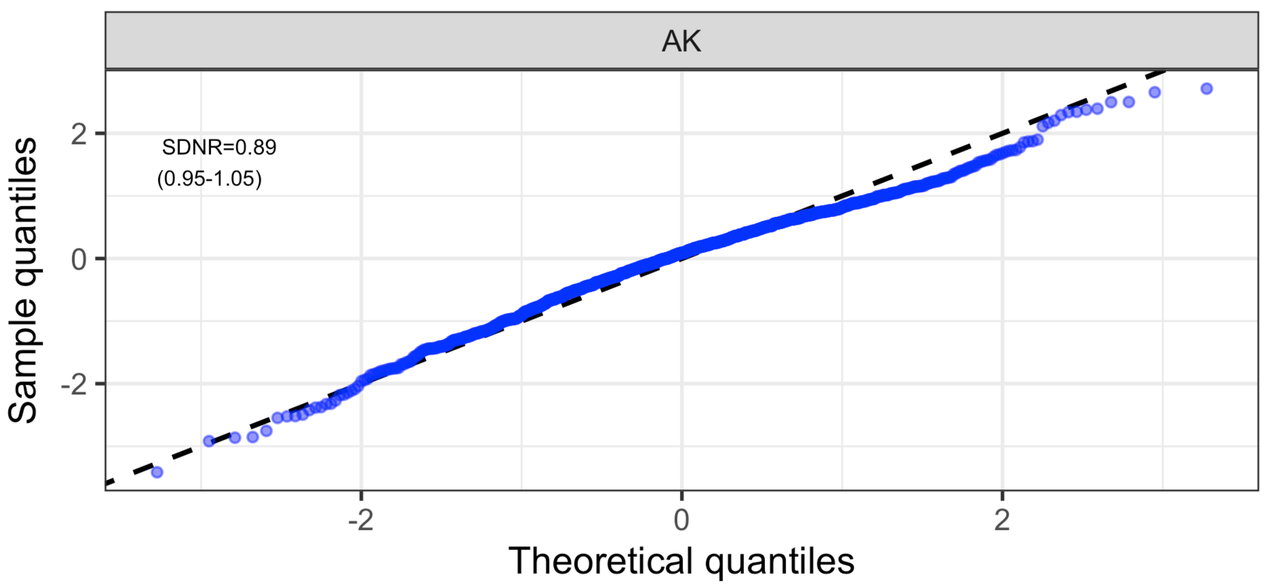
JPN LL Survey Length Comps Male AK

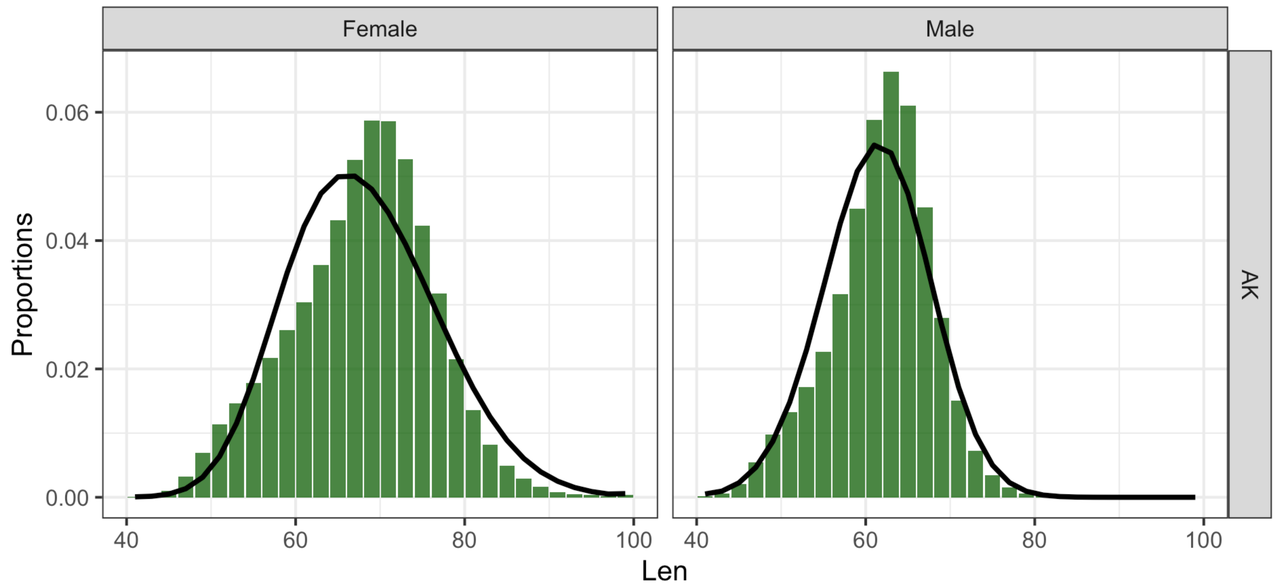
1979 1980 1981 1982
1983 1984 1985 1986
1987 1988 1989 1990
1991 1992 1993 1994

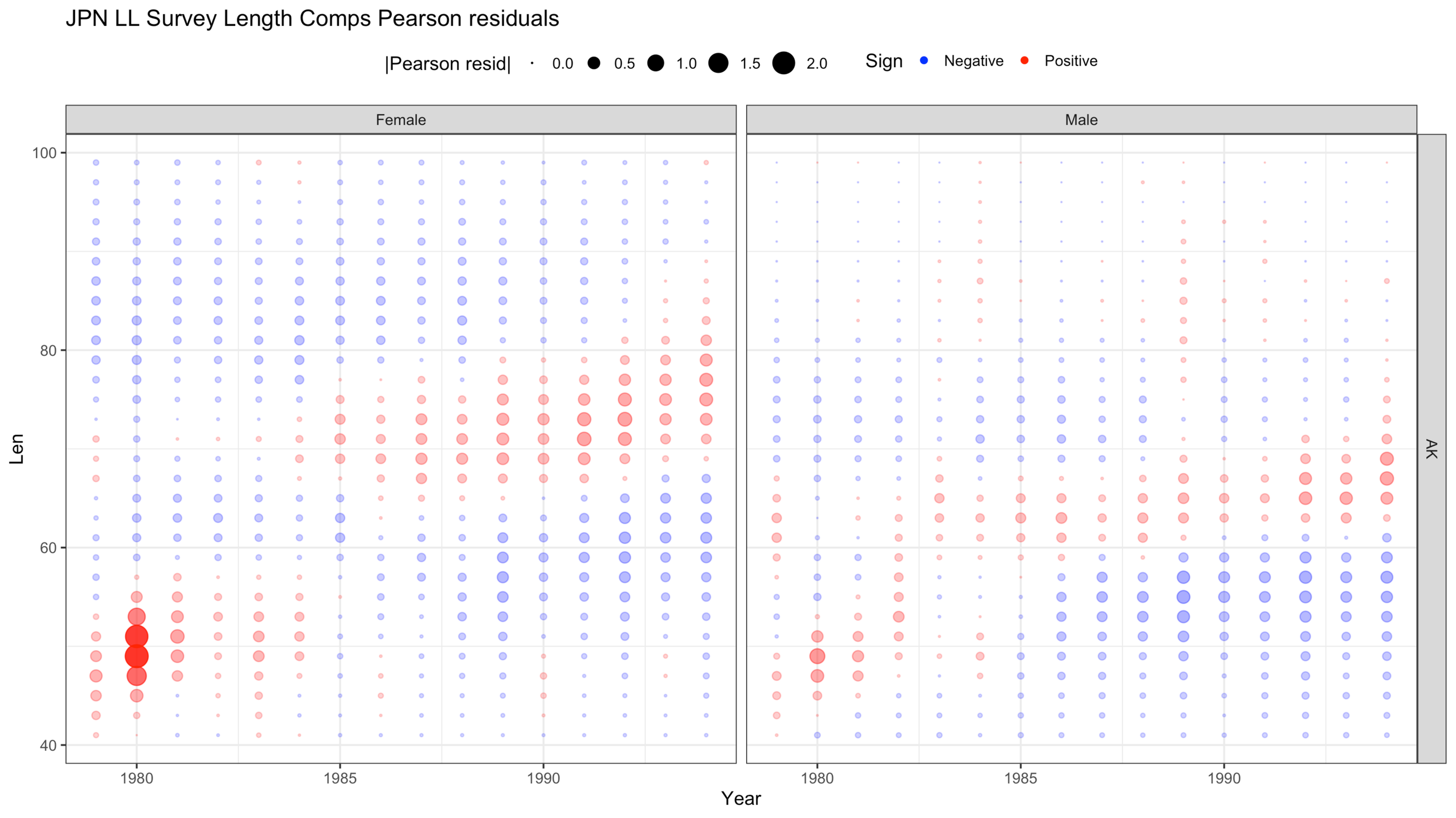
Proportion
0.00 0.02 0.04 0.06 0.08
40 60 80 100
Len

JPN LL Survey Length Comps OSA AK








JPN LL Survey Length Comps Pearson residuals
|Pearson resid|
0.0
0.5
1.0
1.5
2.0
Sign
Negative
Positive
Female
Male
AK
Len
100
80
60
40
1980
1985
1990
Year

Type: Age, Catch, Index, jnLL, Length, Penalty, Prior

26.2_FAA_3LLF_LLS_BS_Dome_Gam

Likelihood

200
0
-200
-400

Catch nLL
Fishery Age nLL
Fishery Length nLL
Fishing Mortality Penalty
Initial Age Penalty
jnLL
M Prior
Recruitment Penalty
Selectivity Penalty
Survey Age nLL
Survey Index nLL
Survey Length nLL

Likelihood Component

Numbers-at-Age

Sex — Female ···· Male

Age- 1 | Age- 2 | Age- 3 | Age- 4 | Age- 5 | Age- 6

Age- 7 | Age- 8 | Age- 9 | Age- 10 | Age- 11 | Age- 12

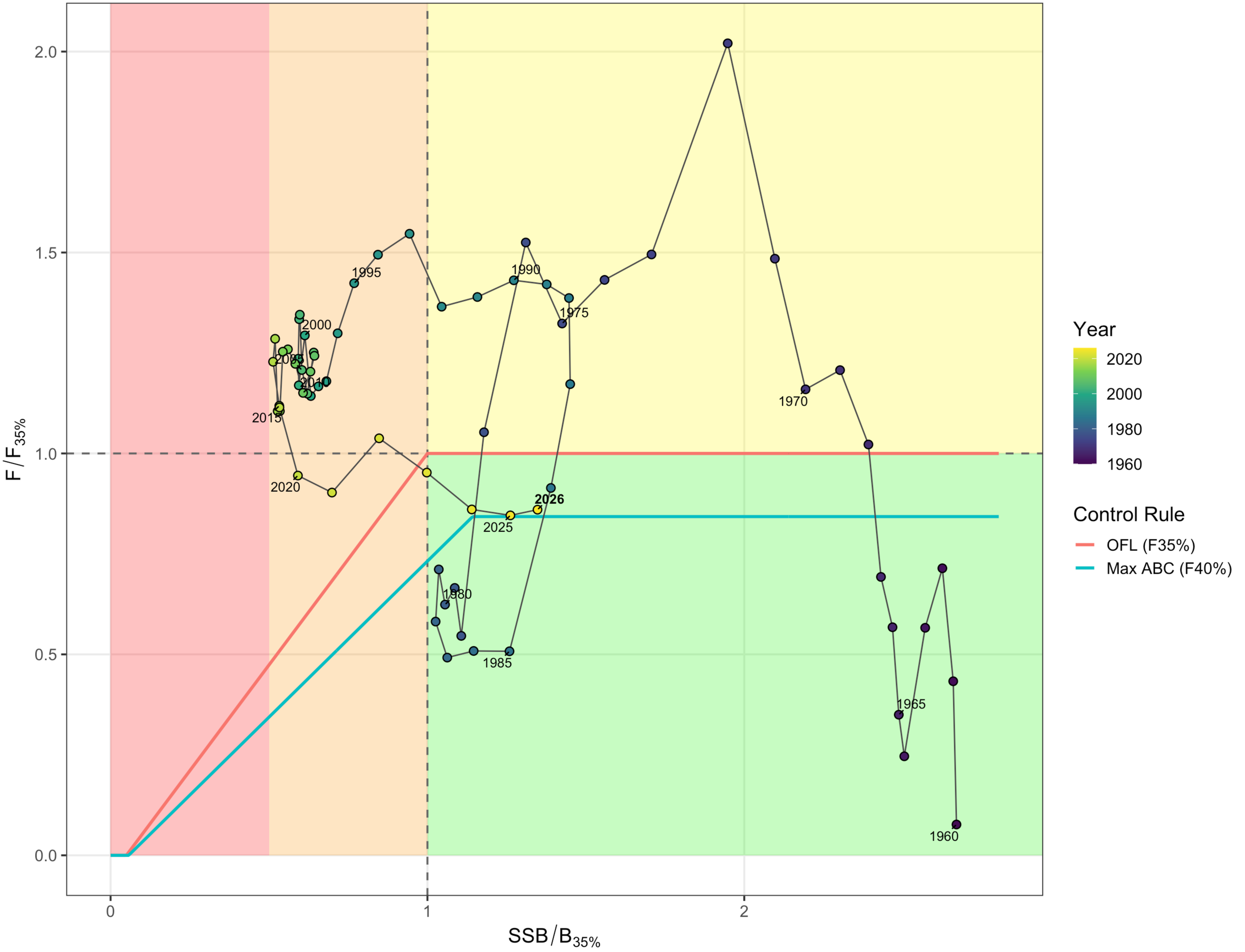
Age- 13 | Age- 14 | Age- 15 | Age- 16 | Age- 17 | Age- 18

Age- 19 | Age- 20 | Age- 21 | Age- 22 | Age- 23 | Age- 24

Age- 25 | Age- 26 | Age- 27 | Age- 28 | Age- 29 | Age- 30

Numbers-at-Age (millions)

Year


2.0
1.5
1.0
0.5
0.0
F/F35%
0
1
2
SSB/B35%
1995
1990
1975
2000
2005
2010
2015
2020
2026
2025
1980
1985
1970
1965
1960
Year
2020
2000
1980
1960
Control Rule
OFL (F35%)
Max ABC (F40%)

Retrospective Year
1 2 3 4 5 6 7

Recruitment

Mohns Rho: -0.094

SSB

Mohns Rho: 0.2713

AK

Relative Difference from Terminal Year

0.5
0.0
-0.5

1960 1980 2000 2020

1960 1980 2000 2020

Year

Peel
1 2 3 4 5 6 7

AK
Recruitment

AK
SSB

Value

100
75
50
25
0

200
100
0

1960
1980
2000
2020

Year

AK

Recruitment (millions)

100
75
50
25

0
1
2
3
4
5
6
7

Years since cohort was last estimated

Cohort
2016
2017
2018
2019
2020
2021
2022
2023

Retrospective Year
1 2 3 4 5 6 7

Recruitment

Mohns Rho: -0.0969

SSB

Mohns Rho: 0.2919

AK

Relative Difference from Terminal Year

1

0

1960 1980 2000 2020

1960 1980 2000 2020

Year

Peel
1 2 3 4 5 6 7

AK
Recruitment

AK
SSB

Value

100
75
50
25
0

200
100
0

1960
1980
2000
2020

Year

AK

Recruitment (millions)

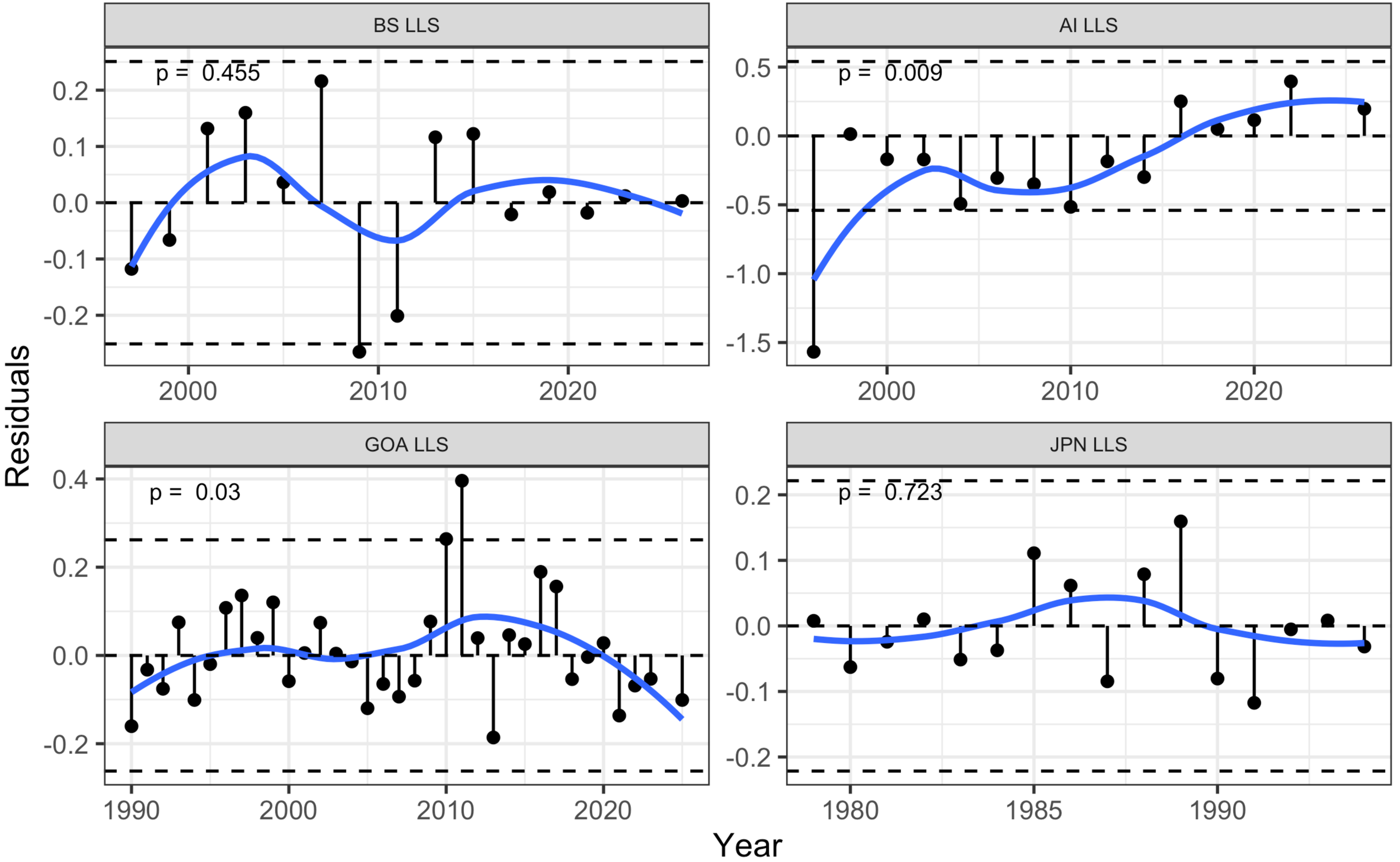
100
75
50
25

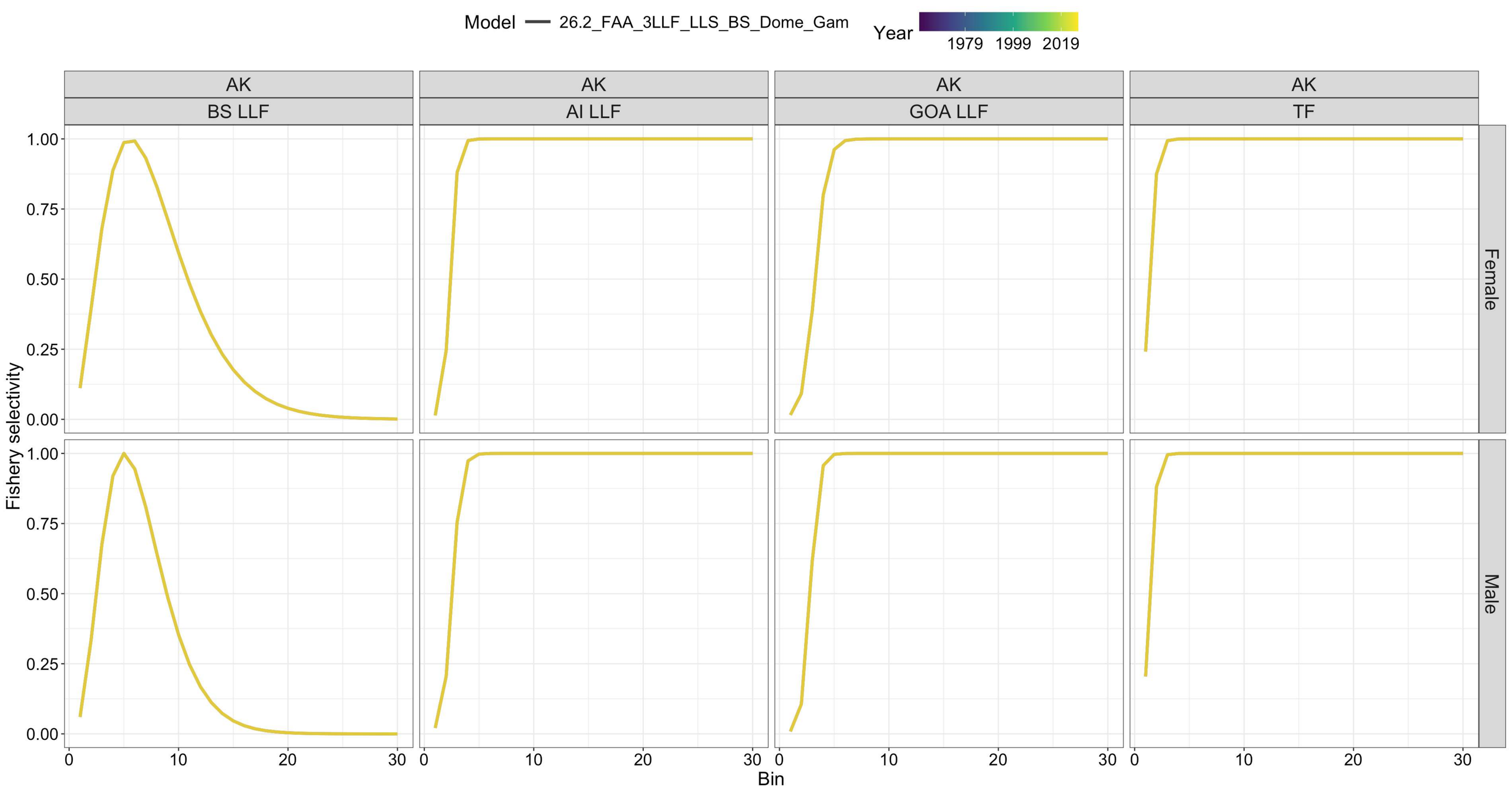
0
1
2
3
4
5
6
7

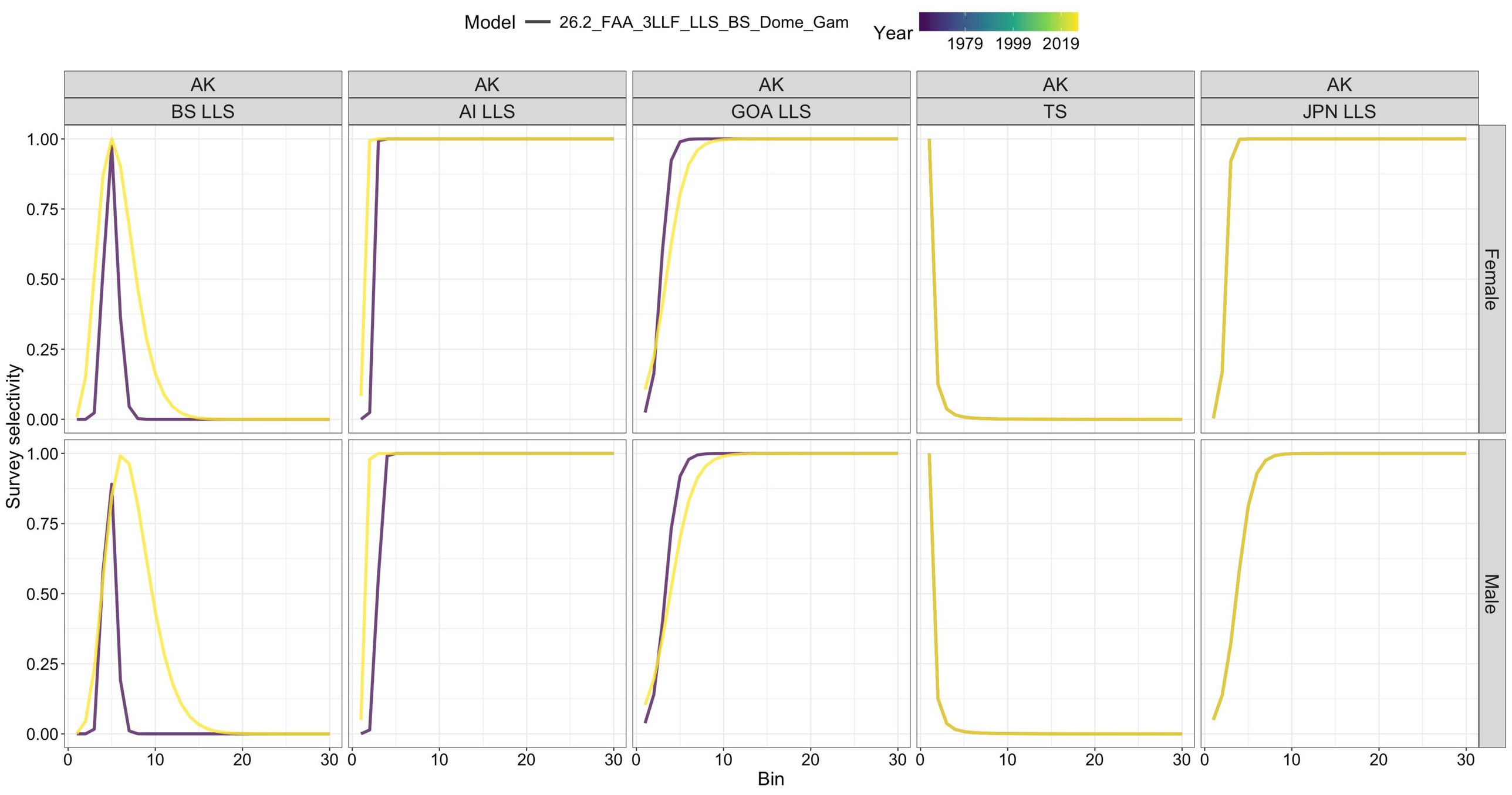
Years since cohort was last estimated

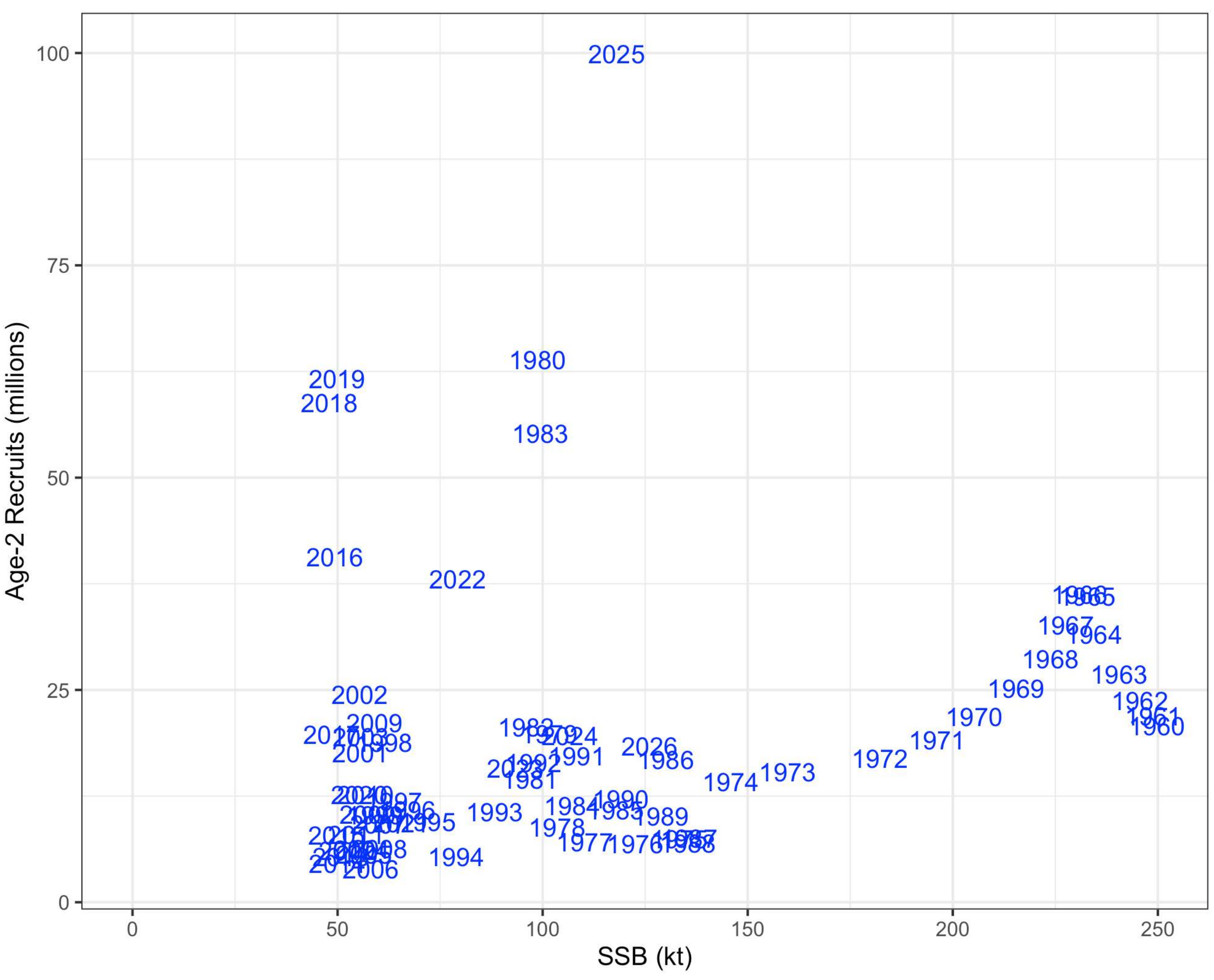
Cohort
2016
2017
2018
2019
2020
2021
2022
2023

Index Residuals Runs Test




Model
26.2_FAA_3LLF_LLS_BS_Dome_Gam
Year
1979
1999
2019
AK
BS LLF
AK
AI LLF
AK
GOA LLF
AK
TF
Female
Male
Fishery selectivity
1.00
0.75
0.50
0.25
0.00
0
10
20
30
Bin


Model
26.2_FAA_3LLF_LLS_BS_Dome_Gam
Year
1979
1999
2019
AK
BS LLS
AK
AI LLS
AK
GOA LLS
AK
TS
AK
JPN LLS
Female
Male
Survey selectivity
1.00
0.75
0.50
0.25
0.00
0
10
20
30
Bin


Age-2 Recruits (millions)
SSB (kt)
0
25
50
75
100
0
50
100
150
200
250
2025
1980
2019
2018
1983
2016
2022
2002
2009
2001
1991
2026
1986
1974
1973
1972
1971
1970
1969
1968
1967
1964
1963
1962
1960
1993
1984
1990
1989
1978
1977
1976
1994
2006

Population and Catch Trends

Recruitment (millions of fish)

100
75
50
25
0

SSB or Catch (kt)

300
200
100
0

1960
1980
2000
2020

Year

Catch
Recruitment
SSB

Model 26.2_FAA_3LLF_LLS_BS_Dome_Gam

AK

Dynamic SSB0

Recruitment

SSB

Total Biom

BS LLF F

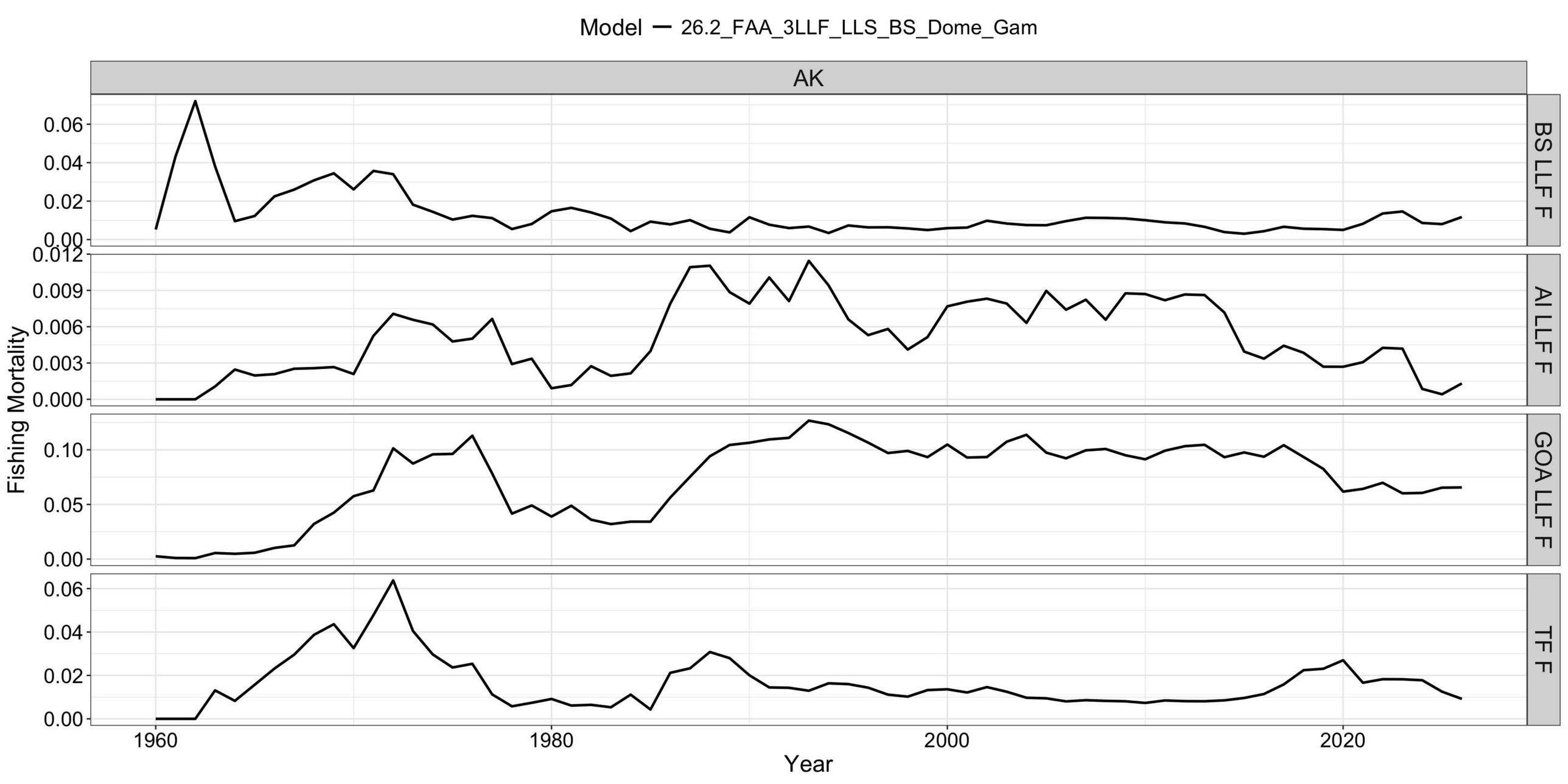
AI LLF F

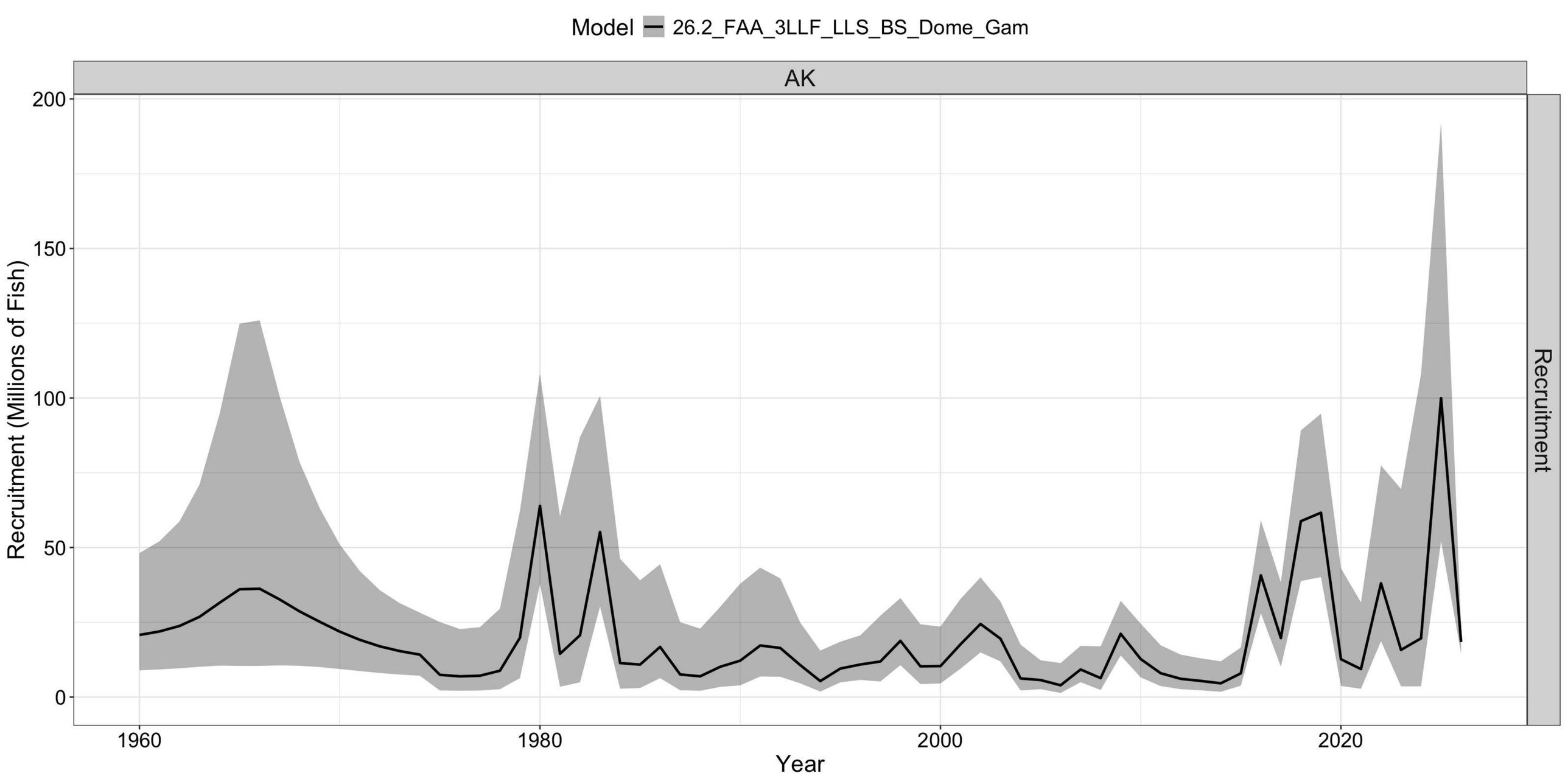
GOA LLF F

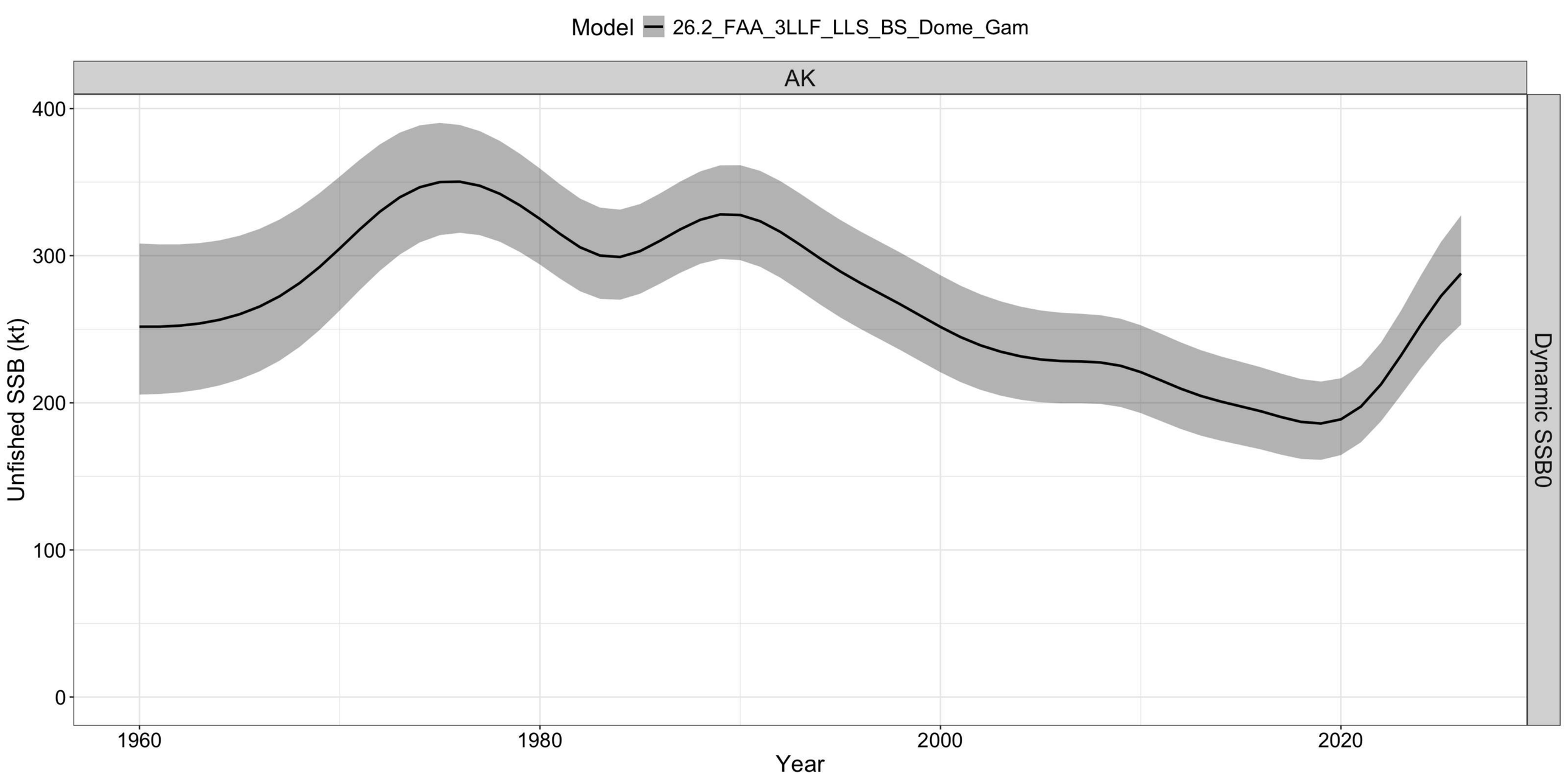
TF F

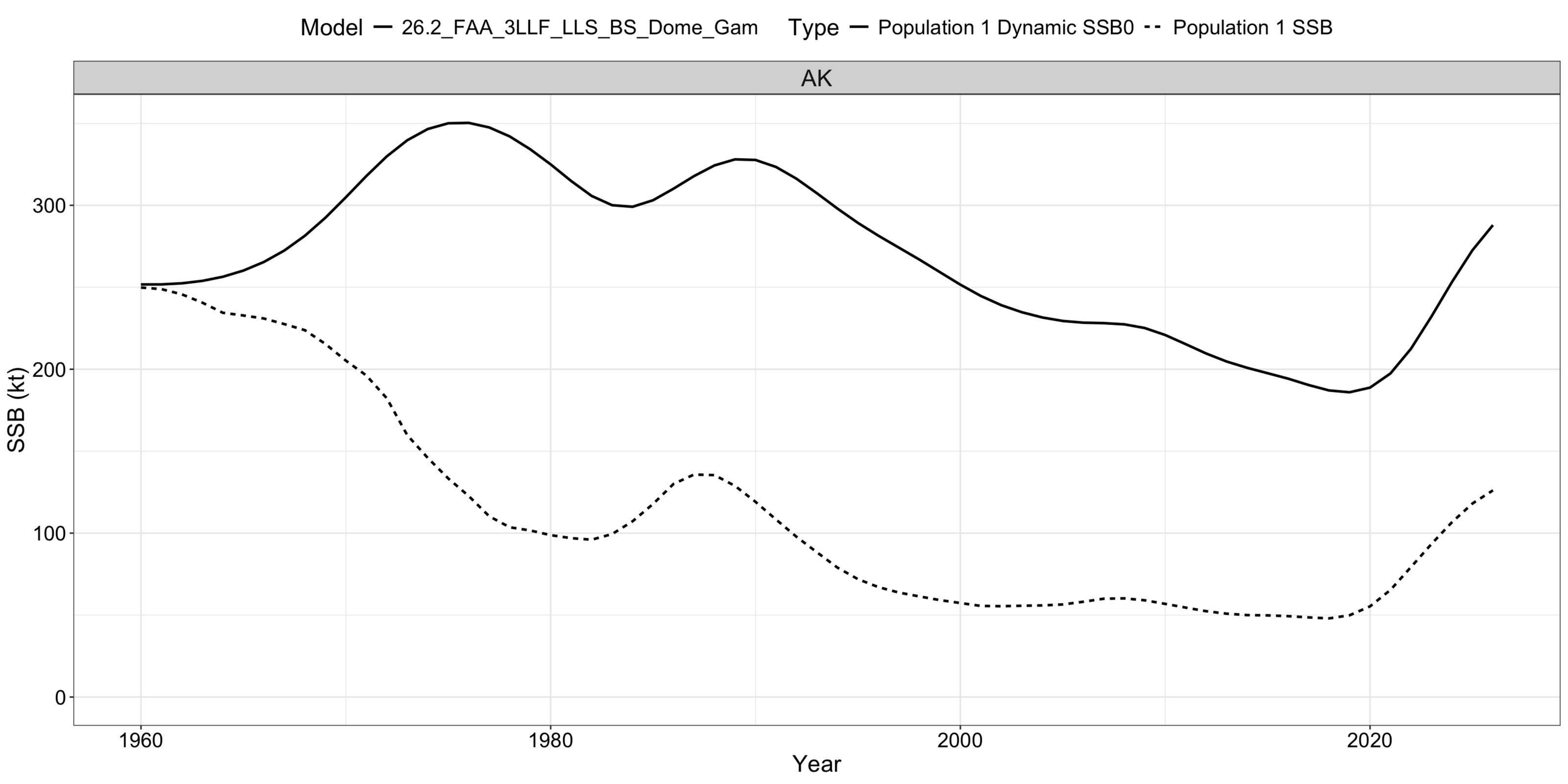
Value

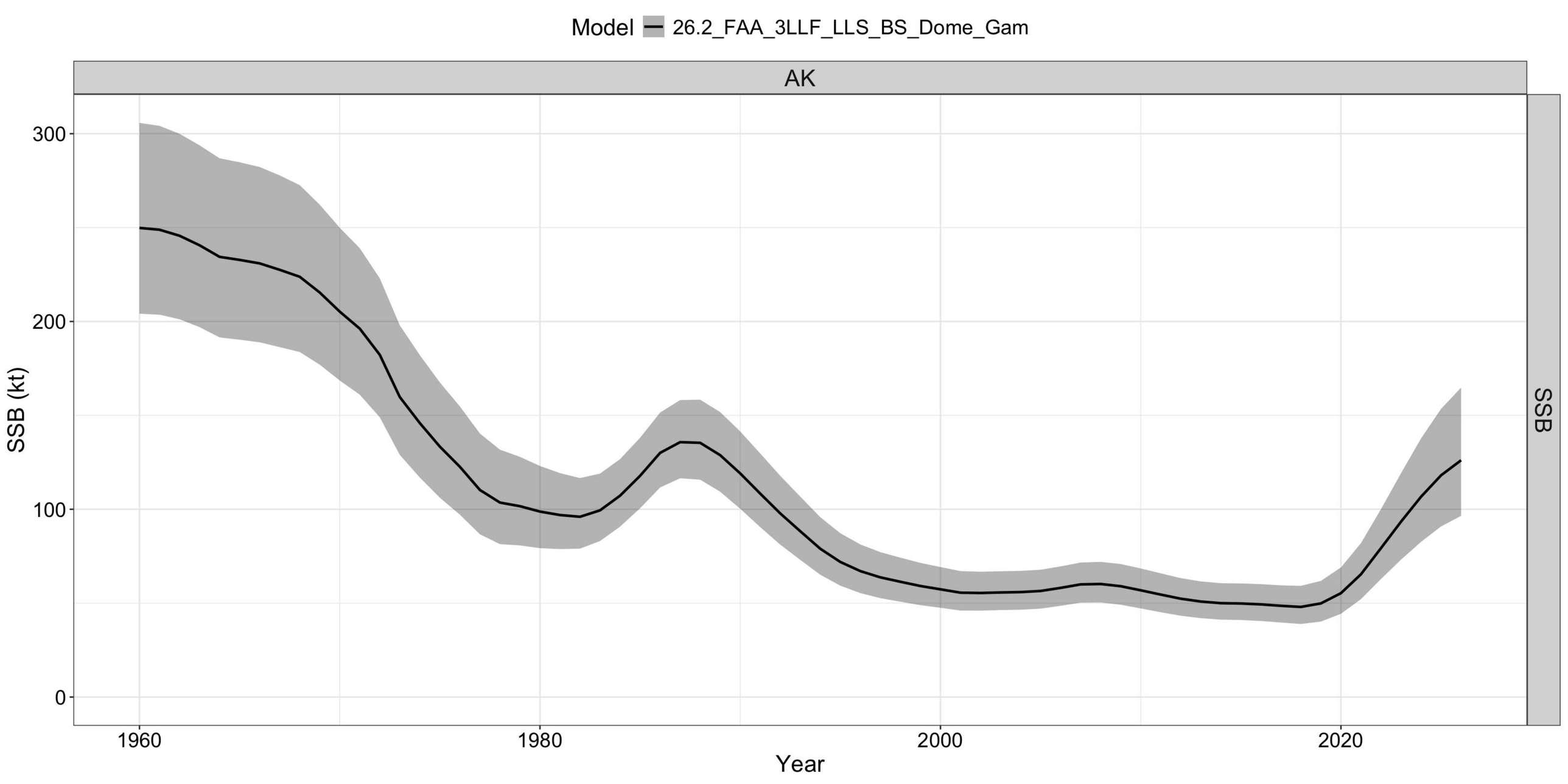
Year

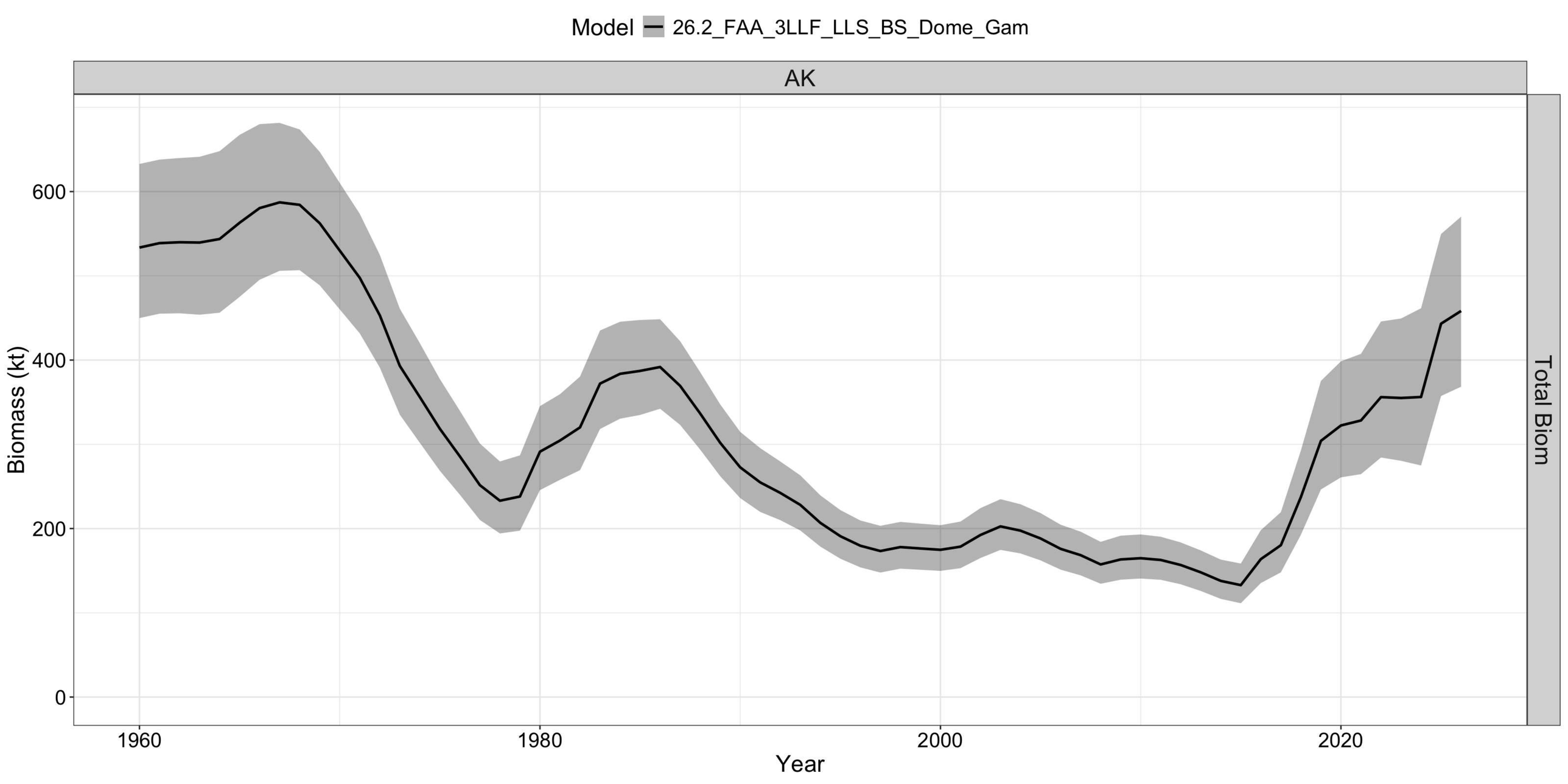

Model — 26.2_FAA_3LLF_LLS_BS_Dome_Gam
AK
BS LLF F
AI LLF F
GOA LLF F
TF F
Fishing Mortality
0.06
0.04
0.02
0.00
0.012
0.009
0.006
0.003
0.000
0.10
0.05
1960
1980
2000
2020
Year


Model
26.2_FAA_3LLF_LLS_BS_Dome_Gam
AK
Recruitment
Recruitment (Millions of Fish)
200
150
100
50
0
1960
1980
2000
2020
Year

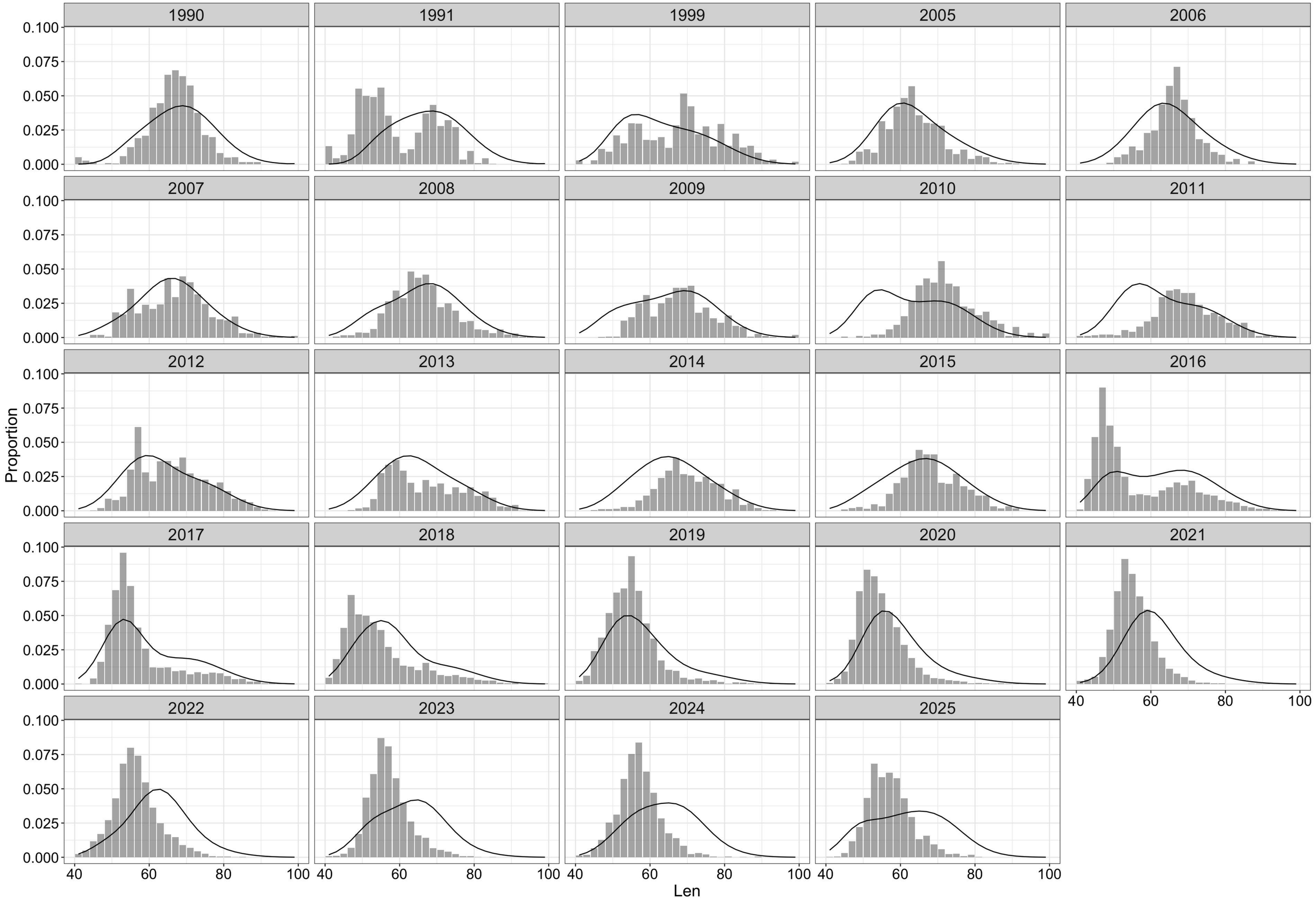

Model
26.2_FAA_3LLF_LLS_BS_Dome_Gam
AK
Dynamic SSB0
Unfished SSB (kt)
400
300
200
100
0
1960
1980
2000
2020
Year


Model — 26.2_FAA_3LLF_LLS_BS_Dome_Gam
Type — Population 1 Dynamic SSB0 -- Population 1 SSB
AK
SSB (kt)
300
200
100
0
1960
1980
2000
2020
Year

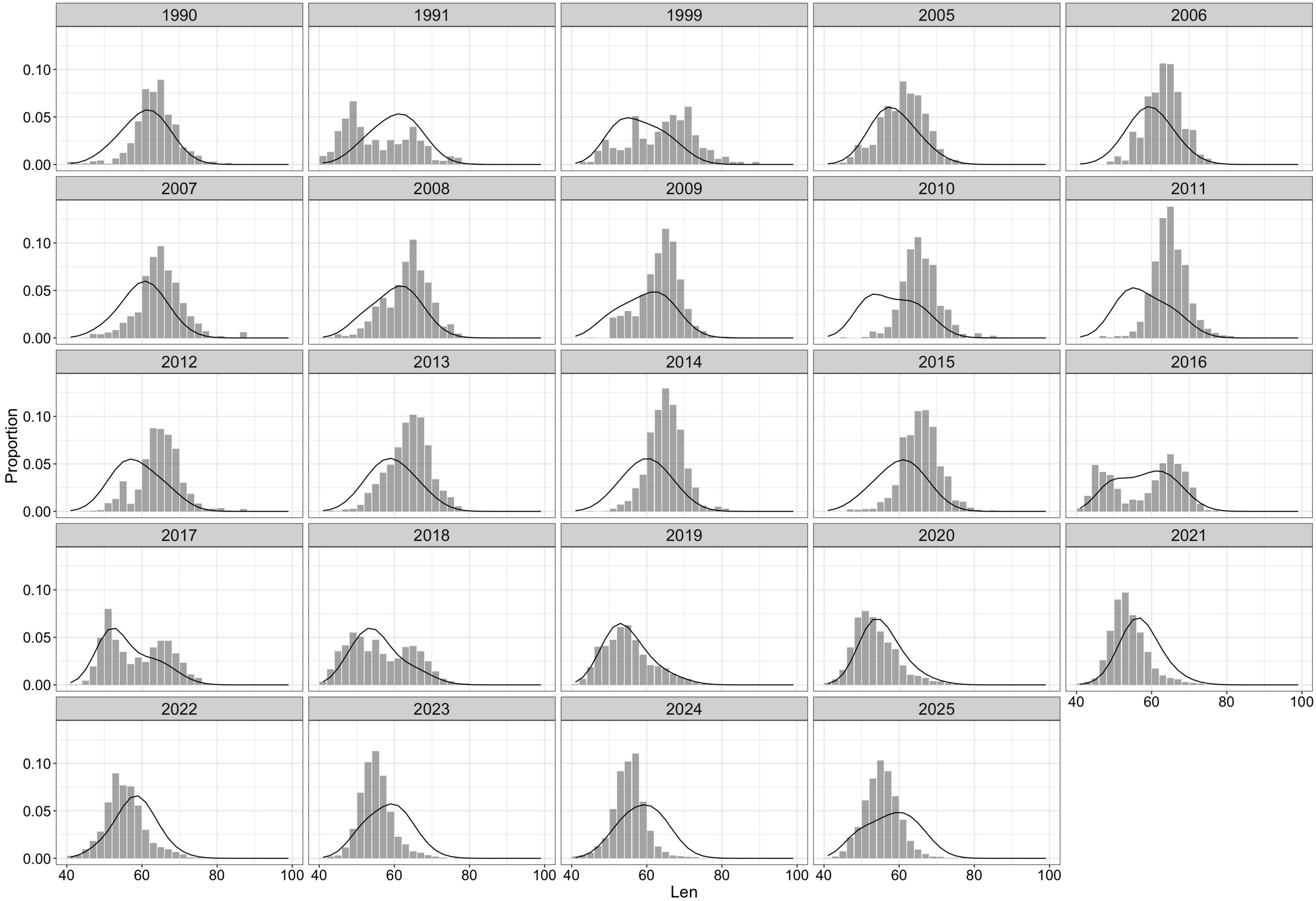

Model
26.2_FAA_3LLF_LLS_BS_Dome_Gam
AK
SSB
SSB (kt)
300
200
100
0
1960
1980
2000
2020
Year


Model
26.2_FAA_3LLF_LLS_BS_Dome_Gam
AK
Total Biom
Biomass (kt)
600
400
200
0
1960
1980
2000
2020
Year

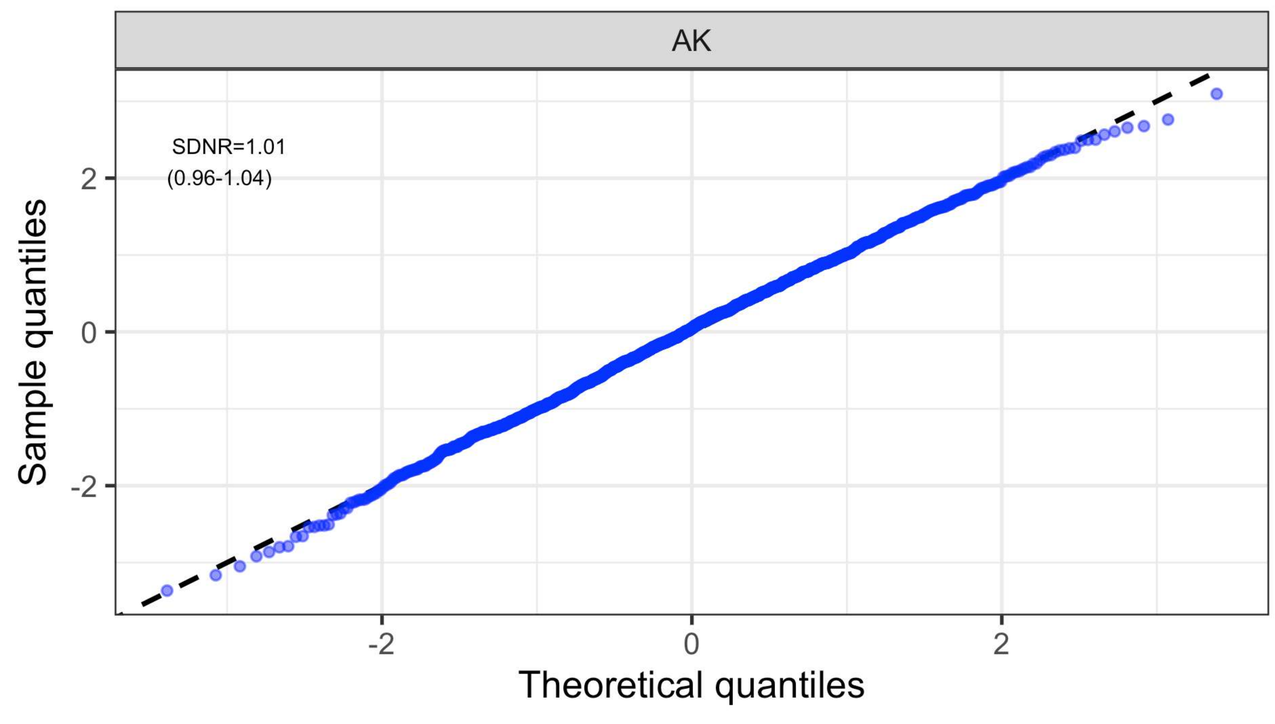
Trawl Fishery Length Comps Female AK

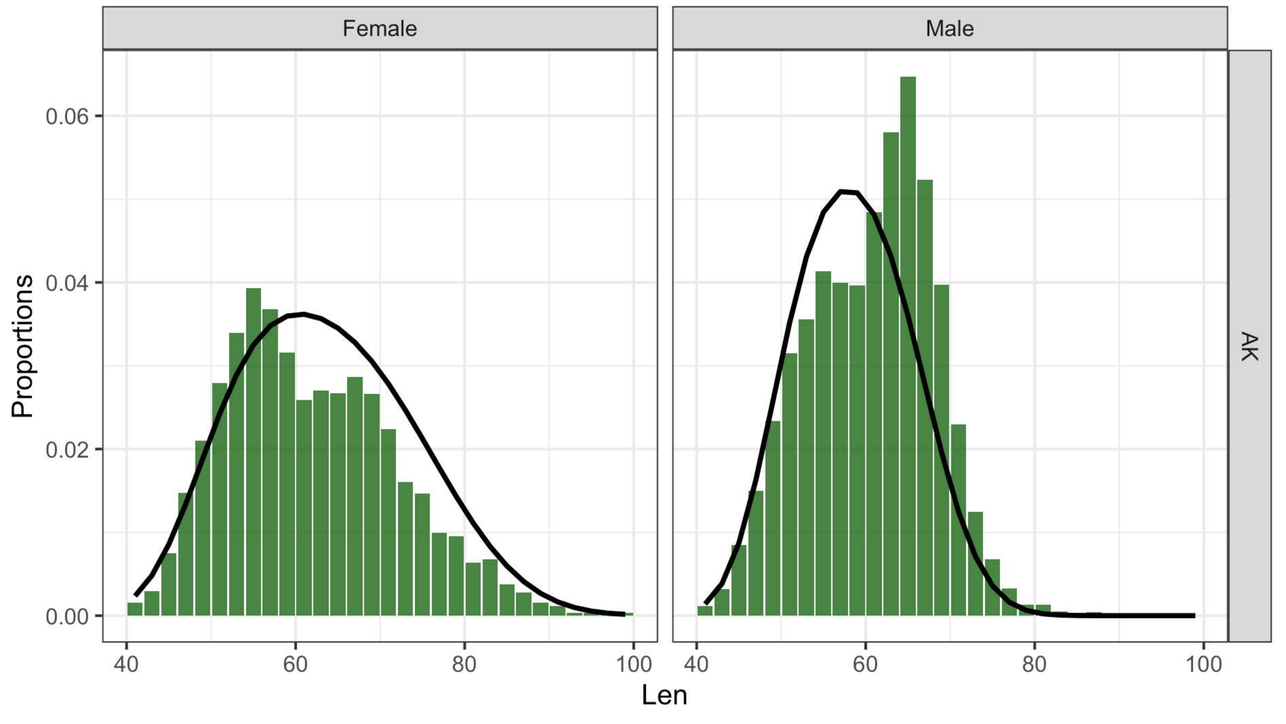


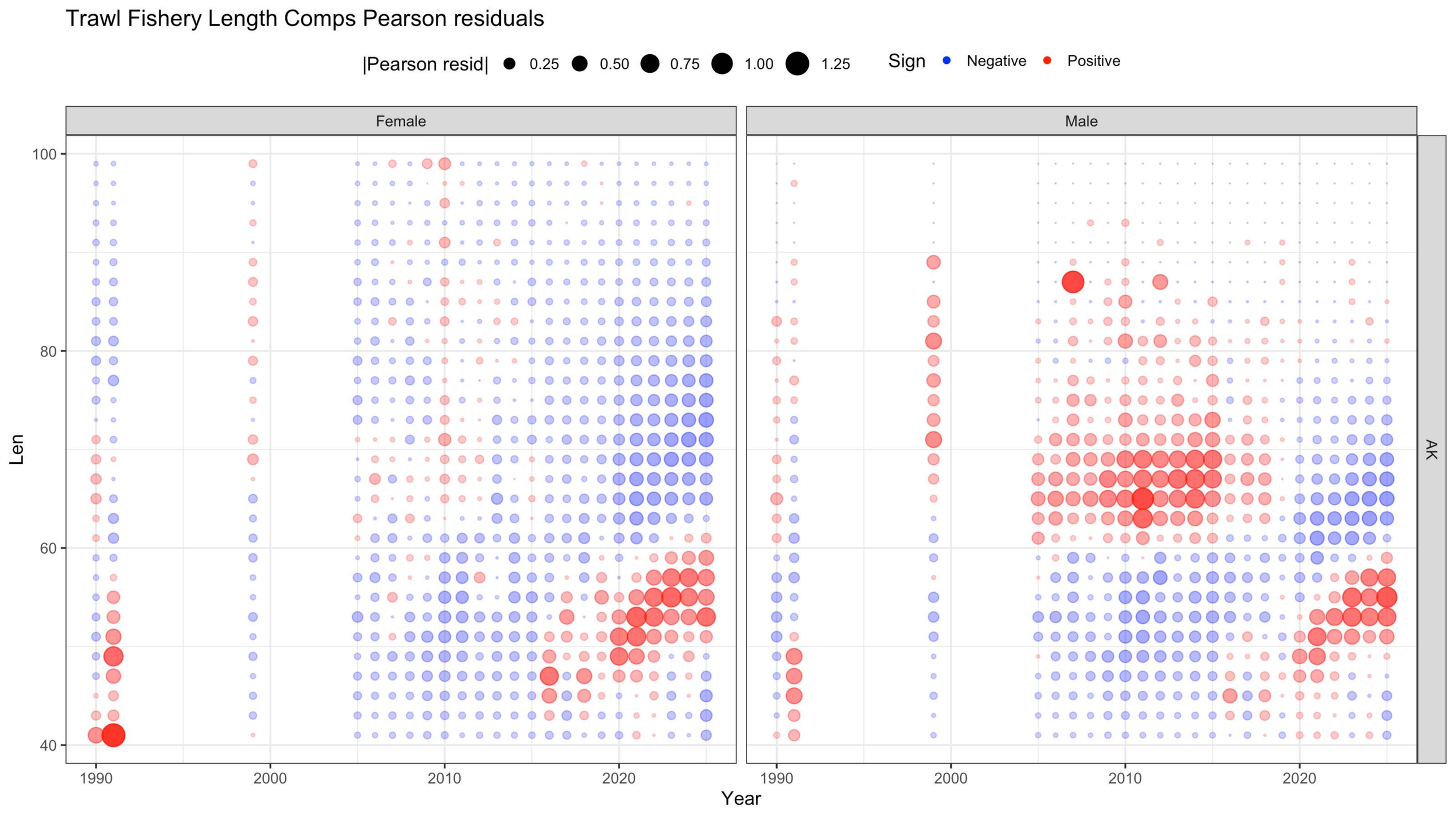
Trawl Fishery Length Comps Male AK



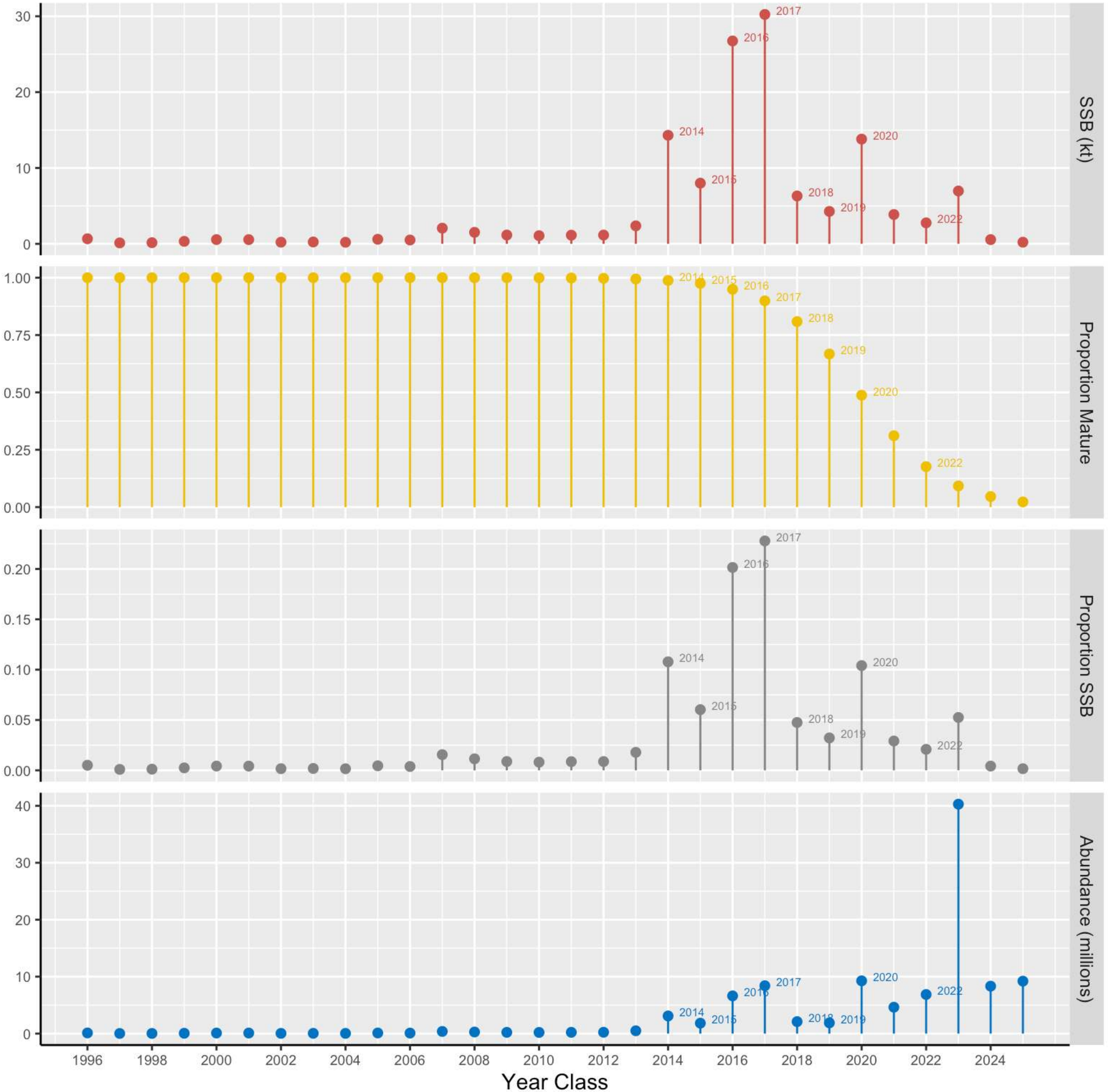
Trawl Fishery Length Comps OSA AK





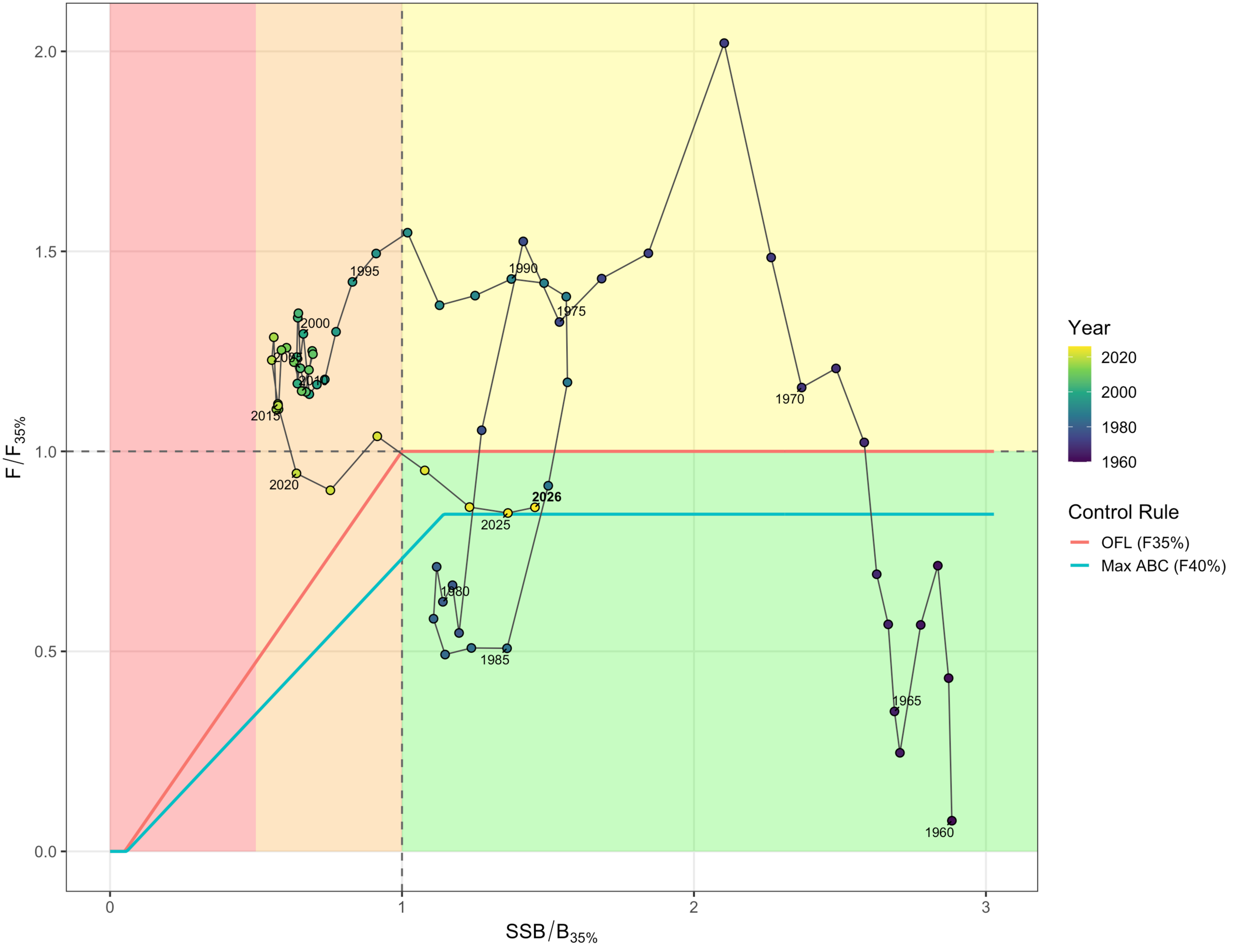

Trawl Fishery Length Comps Pearson residuals
|Pearson resid|
0.25
0.50
0.75
1.00
1.25
Sign
Negative
Positive
Female
Male
AK
Len
100
80
60
40
1990
2000
2010
2020
Year

Contribution to 2027 SSB by Year Class



Source	SE	Data_Wt	ISS	ESS
BS LL Survey Index	0.20	1.00	NA	NA
AI LL Survey Index	0.23	1.00	NA	NA
GOA LL Survey Index	0.08	1.00	NA	NA
Trawl Survey Index	NA	NA	NA	NA
JPN LL Survey Index	0.07	1.00	NA	NA
Fixed Gear Fishery BS CPUE	NA	NA	NA	NA
Fixed Gear Fishery AI CPUE	NA	NA	NA	NA
Fixed Gear Fishery GOA CPUE	NA	NA	NA	NA
BS LL Survey Ages	NA	0.01	34.07	0.34
AI LL Survey Ages	NA	2.10	35.57	74.72
GOA LL Survey Ages	NA	0.35	67.17	23.71
JPN LL Survey Ages	NA	0.18	83.33	14.62
JPN LL Survey Lengths	NA	0.34	54.62	18.49
BS Fixed Gear Fishery Ages	NA	0.09	26.85	2.53
AI Fixed Gear Fishery Ages	NA	0.04	29.65	1.20
GOA Fixed Gear Fishery Ages	NA	0.11	63.44	7.27
Trawl Fishery Lengths	NA	0.10	47.62	5.00

Model	Region	Terminal_SSB	Terminal_SSB0	Terminal_F	Catch_Advice	B_Ref_Pt	F_Ref_Pt	B_over_B_Ref	B_over_DynB_Ref	F_ov
_3LLF_LLS_BS_Dome_Gam	1	126.0951	287.9147	0.08776837	27.90596	106.9437	0.08601	1.17908	0.43796	1


2.0
1.5
1.0
0.5
0.0
F/F35%
0
1
2
3
SSB/B35%
1995
1990
1975
2000
2005
2010
2015
2020
2026
2025
1980
1985
1970
1965
1960
Year
2020
2000
1980
1960
Control Rule
OFL (F35%)
Max ABC (F40%)

Quantity	2025	2026	2027	2028
M Female (estimated)	0.114	0.114	0.096	0.096
M Male (estimated)	0.114	0.114	0.102	0.102
Average Recruitment (millions)	25.9	25.9	17.5	17.5
Tier	3a	3a	3a	3a
Projected Total Biomass (Age-2+, t)	704,713	695,681	464785	476489
Projected Female Spawning Biomass (t)	219,714	241,217	132710	145238
B100% (t)	302,672	302,672	267359	267359
B40% (t)	121,069	121,069	106944	106944
B35% (t)	105,935	105,935	93576	93576
FOFL	0.102	0.102	0.102	0.102
max FABC	0.087	0.087	0.086	0.086
Total Depredation (t)	171	169	131	131
OFL (t)	58,731	57,993	32887	37097
OFLw (t)	58,532	57,797	32756	36966
max ABC (t)	50,283	49,651	27904	31489
ABC (t)	50,283	49,651	27904	31489
ABCw (t)	50,111	49,482	27773	31358
Status	2023	2024	2025	2026
Overfishing	No	NA	No	NA
Overfished	NA	No	NA	No
Approaching Overfished	NA	No	NA	No

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Fishing Mortality

0.00

0.05

0.10

0.15

0.20

1960

1980

2000

2020

2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Fishing Mortality

0.00

0.05

0.10

2010

2020

2030

2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Biomass (kt)

0
200
400
600

1960
1980
2000
2020
2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Biomass (kt)

0

200

400

600

2010

2020

2030

2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Yield (kt)

0
20
40
60

1960
1980
2000
2020
2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Yield (kt)

0
10
20
30

1960
1980
2000
2020
2040

Year

Fleet
BS LLF
AI LLF
GOA LLF
TF

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Yield (kt)

0
10
20
30

2010
2020
2030
2040

Year

Fleet
BS LLF
AI LLF
GOA LLF
TF

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Yield (kt)

0
10
20
30
40

2010
2020
2030
2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Recruitment (millions)

100
75
50
25
0

1960
1980
2000
2020
2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Recruitment (millions)

0

25

50

75

100

2010

2020

2030

2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

SSB (kt)

0

100

200

300

1960

1980

2000

2020

2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

SSB (kt)

0

100

200

300

2010

2020

2030

2040

Year